



REPUBLIC OF KENYA
MINISTRY OF ROADS AND TRANSPORT

PAM 4

Project Appraisal Manual

Part 4: Road Safety Audits

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Foreword

This manual was developed by the Ministry pursuant to The Fourth Schedule of the Constitution which assigns to the National Government the functions and powers of setting standards for the construction and maintenance of all public roads including those under the County Governments.

It is part of a series of manuals that replace the first generation of road manuals developed in the first and second decades after independence. This second generation of the road manuals were developed to cover the entire road project cycle covering planning, appraisal, design, contracts, construction, maintenance, operations and monitoring. The series incorporates best practices, climate change considerations, and recent technologies to enable the provision of road infrastructure that is safe, secure, and efficient.

Under the Kenya Vision 2030 long term plan, infrastructure expansion and modernisation are some of the foundations for the realisation of economic, social and political transformation of Kenya into a rapidly industrialising middle-income country. The plan envisages an integrated, safe and efficient transport and communication infrastructure network consisting of roads, railways, ports, airports, waterways, and telecommunications infrastructure.

The strategies to be pursued under the Vision 2030 plan to improve infrastructure services and to maximise the economic and social impacts of infrastructure development and management include: Strengthening of the institutional framework for infrastructure development and maintenance; Raising efficiency and quality of infrastructure projects; Enhancing local content of identified infrastructure projects to minimise import content; Benchmarking infrastructure facilities and services provision with globally acceptable performance standards; and, Implementing infrastructure projects that will stimulate demand in hitherto marginalised areas.

The first three 5-year Medium Term Plans (MTP) under the Vision 2030 from 2008 to 2022 targeted construction of 1,950 km, 5,500 km and 10,000 km of new paved roads under MTP I, II and III, respectively, totalling 17,450 km. This was a massive infrastructure development program intended to double the paved road network in 10 years compared to 8,600 km developed from independence in 1963 to 2008.

Implementation of MTP I to III resulted in the construction of 14,000 km of paved roads, which extended the paved road coverage to Arid and Semi-Arid regions, that had been previously neglected. However, some key milestones of the Vision 2030 goals have not been realised. This has been due to internal and external challenges. External challenges included: climate change – prolonged droughts; the emergence of COVID-19 pandemic; global supply chain disruptions; exchange rate volatility; and rising interest rates in the leading economies.

The internal challenges included: inadequate road maintenance equipment; pavement overloading by heavy goods vehicles; huge maintenance backlog of the road network; low contracting and supervision capacity particularly in the Counties; poor quality control and assurance of works; congestion in urban areas; encroachment on road reserves; high costs and delays in payments of land acquisition; lack of harmonisation of cross-border transport regulation and operational procedures; rapid urbanisation; increased traffic volume with exponential growth of motorcycle traffic; high cost/delays in relocation of utilities and services along and across road reserves; inadequate funding of projects and programs; and, delay or default in payments for goods, services and works.

The implementation of MTP III came to an end on 30th June 2023, ushering in the implementation of the Fourth Medium Term Plan (MTP IV), which has been aligned to the aspirations of the Kenya Vision 2030 and the Kenya Kwanza Government's Bottom-Up Economic Transformation Agenda (BETA) planning approach and its key priorities.

BETA is the Government's transformation agenda geared towards economic turnaround through a value chain approach. BETA has targeted sectors with the highest impact to drive economic recovery and growth. This will be achieved through bringing down the cost of living; eradicating hunger; creating jobs; expanding the tax base; improving foreign exchange balances; and inclusive growth. BETA ensures rational resource allocation by eliminating wastage of resources occasioned by duplication, overlaps, fragmentation and ineffective coordination in the implementation of programmes and projects.

The Fourth Medium Term Plan key priorities are clustered under five key sectors, namely: Finance and Production; Infrastructure; Social; Environment and Natural Resources; and Governance and Public Administration. The infrastructure sector seeks to: enhance transport connectivity by constructing 6,000 km of new roads, maintaining rural and urban roads, rail, air and seaport facilities and services; expand communication and broadcasting systems; and promote the development of energy generation and distribution by increasing investments in green energy (geothermal, wind, solar and hydro). The infrastructure gap is expected to be bridged by promoting economic participation of the private sector through public private partnerships in the financing, construction, development, operation, and maintenance of infrastructure.

The plan entails a shift of focus to fundamentals in project planning and implementation which include: respect for technical input, regulations and standard practices; adherence to project life cycle i.e., planning, feasibility studies and design before procurement of works; public and stakeholder consultation; procurement within budgetary ceilings; shifting focus during project implementation from the finished product 'black top' to the construction of the foundation; building local capacity particularly MSMEs by ensuring prompt payments; and capacity building at all levels to enable internalisation of policies and processes.

The first generation of the road manuals were used for 35 to 45 years. It is my sincere hope that the second generation of the road standards which have been developed in alignment with the Government's strategy will provide guidance in solving most of the above challenges and those expected to emerge in the next 50 years. Implementation of the manuals will enable achievement of the Government aspirations which include inclusive growth; creation of sustainable employment; building of MSMEs; climate change adaptation and realisation of the UN SDGs; enhanced efficiency in management of infrastructure and transport system; and, laying the foundation for the next national long-term plan at the end of the Vision 2030.

On behalf of the Government of Kenya, I would wish to thank the European Union for financing the development of the first drafts of the manuals in 2009 and the African Development Bank for the financial support in the review and updating of the manuals. I would also like to thank the members of the National Steering Committee and the Technical Task Force for their input. The Technical Administrators, and the Kenya National Highways Authority (KeNHA) for the procurement and able administration of the consultancy Contract. I also thank the Consultant, TRL Limited for their role in providing technical expertise that was essential for the success of the manuals updating exercise. I also wish to express my deepest appreciation to our stakeholders and all those who have contributed to this process and the staff of the Ministry for their continued input.

Hon. Davis K. Chirchir, E.G.H

Cabinet Secretary, Ministry of Roads and Transport

Preface

This Manual is for decision makers, engineers, technicians, consultants, contractors, and concessionaires concerned with road projects, irrespective of category of road or the area where they work. Road safety audit during feasibility study, design, construction, maintenance and operation of roads along with application of safe system approach can reduce crashes on the roadway. The purpose of an audit is to ensure that road users are exposed to minimal risks of crashes on new and existing roads.

This manual has been prepared as a guide for conducting road safety audit of new road designs and existing roads. It contains details about engaging an audit team and the road safety audit process that will assist the members of audit teams to undertake an audit. This manual is a valuable instructional tool for auditors and for Engineers alike.

How to conduct road safety audit is explained along with the key steps. Safe system approach, its principles, and technical tips while auditing are explained.

A draft Terms of Reference has been provided to assist in selection of a team for conduction of audit by independent auditors. Risk assessment of safety concerns is elaborated along with suggested recommendations.

The manual includes audit case studies from typical road projects. Important safety findings have been presented for different stages of audit.

To further assist audit teams, a full set of audit checklists is included in this manual. There is one checklist for each stage of audit. These checklists prompt audit teams to consider the multitude of safety issues that can arise in road projects including the safety needs of vulnerable road users. The operational issues of managing road safety audit are a part of Roads Manual Volume A: General - Procedures and Standards, to be prepared.

Eng. Joseph M. Mbugua, CBS

Principal Secretary, State Department for Roads

Document Management

Document Status:

This document has the status of a Manual. Users shall apply the contents there-in to fully satisfy the requirements set out. The content of the manual is based on current practice in Kenya and latest practices in the road sector, both regionally and internationally.

Sources of the Document:

Copies of the document can be obtained from:

The Principal Secretary, State Department for Roads, Ministry of Roads and Transport, Works Building, Ngong Road, P.O. Box 30260 - 00100, NAIROBI Email: ps@road.go.ke

A secured PDF copy maybe downloaded from: www.roads.go.ke/downloads

Notification of Errors and Requests for Amendments:

While all care and consideration has been applied in the compilation of this document, the Ministry accepts no responsibility for failure in any way related to the application of this manual or any reference documents cited in it.

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A National Steering Committee was set up and chaired by the Permanent Secretary, Ministry of Roads and Transport, with the following membership: Principal Secretary for Devolution, Office of the Deputy President; Chief Executive Officer, Inter-Governmental Relations Technical Committee; Chief Executive Officer, Council of Governors; Managing Director and Council Secretary, Kenya Bureau of Standards; Director, National Transport and Safety Authority; Director General, Kenya Roads Board; Director General, Kenya Wildlife Services; Chief Executive Officer, Engineers Board of Kenya; Director General, Kenya Rural Roads Authority; Director General, Kenya Urban Roads Authority; President, Institution of Engineers Kenya; Director Policy, Strategy and Compliance; Kenya National Highways Authority; Chief Engineer, Roads Division, State Department for Roads; Chief Engineer, Materials Testing and Research Division, State Department for Roads.

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Abbreviations

AIP	Accident Investigation and Prevention Techniques
CAR	Corrective Action Report
GDP	Gross Domestic Product
HSE	Health Safety and Environment
iRAP	International Road Assessment Programme
NMT	Non-Motorised Traffic
PCU	Passengers Car Units
PIARC	World Road Association
PPE	Personal Protective Equipment
PwD	Persons with Disabilities
RoW	Right of Way
RSA	Road Safety Audit
RSE	Road Safety Engineering
TMP	Traffic Management Plan
USD	United States Dollar
VMS	Variable Message Sign
VRU	Vulnerable Road User

Glossary of Terms

6C	Road signs that are Clear, Conspicuous, Comprehensive, Credible, Consistent and Correct.
Access Control	The condition where the road agency controls the right of landowners to direct access to and from a public highway.
Audit brief	The instructions to the Audit Team defining the scope and details of the road or development project to be audited, including sufficient information for the audit to be undertaken.
Audit Team Continued Professional Development	A record of courses, conferences, workshops, and any other training undertaken with the purpose of keeping Road Safety Auditors up to date with the latest developments in Road Safety Audit, crash investigation, crash data analysis and Road Safety Engineering.
Authority	A government organisation in Kenya that is responsible for a specific area or function and with vested power or right to make decisions or enforce regulation. These include Ministries, Departments and Agencies (MDA) at national and county levels.
Auxiliary Lane	Part of the roadway adjoining the carriageway for parking, speed change, turning, storage for turning, weaving, truck climbing, and for other purposes supplementary to through traffic movement.
Bus Lay-bys	Lay-by reserved for public service vehicles.
Camber	The convex shape given to the curved cross-section of a roadway.
Capacity	The maximum sustainable flow rate at which vehicles or persons can reasonably be expected to traverse a point or uniform segment of a lane or roadway during a specified time period under given roadway, geometric, traffic, environmental, and control conditions; usually expressed as vehicles per hour, passenger cars per hour, or persons per hour.
Carriageway	Portion of the roadway including the various physically contiguous traffic lanes and auxiliary lanes, serving one or both directions of traffic, and not including shoulders.
Channelisation	The use of pavement markings or islands to direct traffic through an intersection.
Clear Zone	Unencumbered roadside recovery area.
Climbing Lane	An auxiliary lane in the upgrade direction for use by slow moving vehicles and to facilitate overtaking, thereby maintaining capacity and freedom of operation on the carriageway.
Corrective Action Report	A tabular summary report prepared by the Audit Team to be completed by the client to respond to the identified findings and recommendations detailed in the audit report.
Crash Investigation	The on-site investigation and collection of evidence to determine the events and circumstances which contributed to the occurrence of a crash and / or to the severity of the outcome.
Crest	Peak formed by the junction of two gradients.
Cross fall	The transverse slope with respect to the horizon, measured in % transversely across the surface of the roadway.
Culvert	A structure, usually for conveying water under a roadway but can also be used as a pedestrian or stock crossing, with a clear span of less than six metres.

Glossary of Terms *(continued)*

Cycle Lane	A portion of the roadway which has been designated by road markings, striping and signing as being exclusively for the use of cyclists.
Cycle Path	A path physically separated from motorised traffic by an open space or barrier and located either within the road reserve or an independent reserve.
Deceleration Lane	An auxiliary lane to enable a vehicle leaving the through traffic stream to reduce speed without interfering with other traffic.
Design Speed	An index which links road function, traffic flow and terrain to the design parameters of sight distance and curvature to ensure that a driver is presented with a reasonably consistent speed environment. It is now defined as the 85th percentile speed of passenger cars travelling in free flow conditions. In practice, most roads will only be constrained to minimum parameter values over short sections or on specific geometric elements.
Diverging	The opposite of merging. When a traffic stream splits into two or more streams.
Expressway	A multilane, divided highway with a minimum of two lanes for the exclusive use of traffic in each direction and control of access without traffic interruption.
Footway	The rural equivalent of the urban footpath.
Gore Area	The triangular plot of land designated when a road forks at the intersection with a second road, or merges on/off from a larger one.
Grade Separator	A structure that makes crossing of two roads, or a road and a railway, at different levels possible.
Gradient	Rate of rise or fall on any length of road, with respect to the horizontal. It is typically expressed as a percentage or as the vertical rise or fall in m/100 m. In the direction of increasing stake value, upgrades are taken as positive and downgrades as negative.
High Speed	Speeds above 80km/h.
Horizontal Clearance	Lateral clearance between the edge of shoulder and obstructions.
Interchange	A system of interconnecting roads (referred to as ramps) in conjunction with one or more grade separations providing for the movement of traffic between two or more roadways which are at different levels at their crossing point.
Kerb	Concrete, often precast, element adjacent to the carriageway and used for drainage control, delineation of the pavement edge or protection of the edge of surfacing. Usually applied only in urban areas.
Median	The area in the middle of a roadway separating the opposing traffic flows. The median thus includes the inner shoulders.
Merging	Movement of a vehicle or vehicles from one or more lanes into a traffic stream.
Operating Speed	Highest overall speed at which a driver can travel on a given road under favourable weather conditions and under prevailing traffic conditions without at any time exceeding the safe speed as determined by the design speed on a section-by-section basis, neither exceeding at any time the speed limit.
Overpass	Grade separation where a road passes over an intersecting road or railway.

Glossary of Terms *(continued)*

Pedestrian Refuge	Raised platform or a guarded area so sited in the roadway as to divide the streams of traffic and to provide a safe area for pedestrians.
Pocketing	When a vehicle hits a guard rail, the rail deflects outwardly. If there is large spacing between guard rail posts, a localised bulging of the rail occurs which is called pocketing.
Ramp	A one-way, often single-lane, road providing a link between two roads that cross each other at different levels.
Right Turn Lane	Auxiliary lane to accommodate deceleration and storage of right- turning vehicles at junctions.
Road Authority	The Highways Authority, the Rural Roads Authority or the Urban Roads Authority in terms of the Kenya Roads Act, 2007.
Roadside	A general term denoting the area beyond the shoulder breakpoints.
Road Safety Audit	A formal systematic road safety assessment of a road project carried out by an independent, qualified auditor who reports on the project's accident potential for all kind of road users.
Road Safety Inspection	Road Safety Inspection (RSI) is an on-site systematic review of an existing road section to identify hazardous conditions, faults or deficiencies that may lead to road crashes.
Roadway	The area normally travelled by vehicles and consisting of traffic lanes, including auxiliary lanes and shoulders.
Rural Road or Highway	Characterised by low volume high-speed flows over extended distances. Usually without significant daily peaking but could display heavy seasonal peak flows.
Sag Curve	Concave vertical curve.
Shoulder	Part of the road outside the carriageway, but at substantially the same level, for accommodation of stopped vehicles for emergency use, for lateral support of the carriageway and for use by pedestrians and cyclists when no other facility has been provided.
Side Drain	Open longitudinal drain situated adjacent to and at the bottom of cut or fill slopes.
Side Slope	Area between the outer edge of shoulder or hinge point and the ditch bottom.
Sight Distance	Distance visible to the driver of a passenger car measured along the normal travel path of a roadway to the roadway surface or to a specified height above the roadway surface, when the view is unobstructed by traffic.
Standard	Design value that may not be transgressed, e.g. an irreducible minimum or an absolute maximum, except in unusual conditions and with the Ministry's approval. For geometric design, a 'standard' is not to be understood as an indicator of quality, i.e. an ideal to be strived for.
Stopping Sight Distance	Distance required by a driver of a vehicle, travelling at a given speed, to bring his vehicle to a stop after an object on the roadway becomes visible. It includes the distance travelled during the perception and reaction times and the vehicle braking distance.

Glossary of Terms *(continued)*

Super-elevation	Inward tilt or transverse inclination given to the cross section of a carriageway throughout the length of a horizontal curve to reduce the effects of centrifugal force on a moving vehicle; expressed as a percentage.
Taper	Slow increase or decrease of width between lane addition or lane drop.
Traffic Flow	Number of vehicles or persons that pass a specific point in a stated time, in both directions unless otherwise stated.
Traffic Lane	Part of a carriageway intended for a single stream of traffic in one direction, which has normally been demarcated as such by road markings.
Transition Curve	Curve whose radius changes continuously along its length, used to connect a tangent with a circular arc or two circular areas of different radii.
Turning Lanes	Lanes which separate turning vehicles from the through traffic lanes.
Underpass	A grade separation where the subject highway passes under an intersecting highway.
Visual Deceit	An optical effect where a driver misunderstands the road ahead.
Vertical Alignment	Direction of the centreline of a road in side profile.
Vertical Curve	Curve on the longitudinal profile of a road, normally parabolic.
Visibility Splay	A drawing plan than visualises the angle and distance from which drivers emerging from an access can see and be seen by drivers proceeding along the priority road. It ensures that entrances have good sight triangle to prevent crashes.
Walkway	That portion of a road reserved exclusively for pedestrians. Often referred to as footpath.
Weaving	Movement in the same general direction of vehicles within two or more traffic streams intersecting at a shallow angle so that the vehicles in one stream cross other streams gradually.

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1 Introduction

1.1 General

This manual was prepared by the Ministry of Roads and Transport as part of the Roads Manual that attends to the project cycle for the provision, maintenance, and operation of roads in Kenya.

The manual is applicable to roads at national and county levels and incorporates best practices, climate change considerations, and modern technologies thereby enabling the provision of road infrastructure that is safe, secure and efficient.

The Kenya road manual series is as follows:

Project Cycle Stage	Manual: Volume or Part/Chapter	Code
A. General	Procedures and Standards Manual	PSM
	1. General	
	2. Policies	
	3. Procedures Guidance	
	4. Codes of Practice	
	5. Guidelines	
B. Planning	Network and Project Planning Manual	NPM
	1. Road Classification	
	2. Route/Corridor Planning	
	3. Route/Corridor Planning	
	4. Highway Capacity	
	5. Project Planning	
C. Appraisal	Project Appraisal Manual	PAM
	1. Environmental Impact Assessment and Audit	
	2. Social Impact Assessment	
	3. Traffic Impact Assessment	
	4. Road Safety Audit	
	5. Project Appraisal	
D. Design	Road Design Manual	RDM
	1. Geometric Design	
	2. Hydrology and Drainage Design	
	3. Materials and Pavement Design for New Roads	
	4. Bridges and Retaining Structures Design	
	5. Pavement Maintenance, Rehabilitation and Overlay Design	
	6. Traffic Control Facilities and Communication Systems Design	
	7. Road Lighting Design	
E. Contracts	Works and Services Contracts Manual	WSCM
	1. Forms of Contracts	
	2. Standard Specification for Road and Bridge Construction	
	3. Bills of Quantities	
F. Construction	Road Construction Manual	RCM
	1. Construction Management	
	2. Project Management	
	3. Site Supervision	
	4. Quality Assurance	
	5. Quality Control	

This table continues onto the next page...

Project Cycle Stage	Manual: Volume or Part/Chapter	Code
G. Maintenance	Road Asset Management Manual	RAAM
	1. Maintenance Management	
	2. General Maintenance	
	3. Pavement Maintenance	
	4. Bridges & Structures Maintenance	
H. Operations	Road Operation Manual	ROM
	1. Traffic Management	
	2. Vehicle Load Control	
	3. Emergency Services	
	4. Tolling	
I. Monitoring & Evaluation	Road Design Manual	MEM
	1. Performance Monitoring Manual	
	2. Technical Audits	
	3. Poverty, Gender Equality and Social Inclusion Monitoring	

This Road Safety Audit Manual is a part of Project Appraisals Manual, Part 4. The operational issues of managing road safety audit are a part of Roads Manual Volume A: General - Procedures and Standards, to be prepared. In order to implement a road safety audit in an effective manner, it is important to develop the Procedures and Standards Manual to guide the process of auditing.

This Manual should be read in conjunction with the Road Design Manual Volume 1 – Part 3, Geometric Design of Highways, Rural and Urban Roads. The complete Roads Design Manual (RDM) is made up of a series of volumes, parts and coding system as shown in [Table 1.1](#).

Table 1.1 Road Design Manual (RDM) Coding

Vol.	Manual Title	Part Name	Code
1	Road Design Manual: Vol. 1 Geometric Design	Part 1 – Topographic Survey	RDM 1.1
		Part 2 – Traffic Surveys	RDM 1.2
		Part 3 – Geometric Design of Highways, Rural and Urban Roads	RDM 1.3
2	Road Design Manual: Vol. 2 Hydrology & Drainage Design	Part 1 – Hydrological Surveys	RDM 2.1
		Part 2 – Drainage Design	RDM 2.2
3	Road Design Manual: Vol. 3 Materials & Pavement Design for New Roads	Part 1 – Ground Investigations and Material Prospecting	RDM 3.1
		Part 2 – Materials Field and Laboratory Testing	RDM 3.2
		Part 3 – Pavement Foundation and Materials Design	RDM 3.3
		Part 4 – Flexible Pavement Design	RDM 3.4
		Part 5 – Rigid Pavement Design	RDM 3.5
4	Road Design Manual: Vol. 4 Bridges & Retaining Structures Design	Part 1 – Geotechnical Investigation and Design	RDM 4.1
		Part 2 – Bridge and Culvert Design	RDM 4.2
		Part 3 – Retaining Structures Design	RDM 4.3
		Part 4 – Reinforced Fill Structures Design	RDM 4.4
		Part 5 – Bridges and Structures Condition Survey	RDM 4.5
		Part 6 – Bridge Maintenance Design	RDM 4.6
5	Road Design Manual: Vol. 5 Pavement Maintenance, Rehabilitation & Overlay Design	Part 1 – Pavement Condition Survey	RDM 5.1
		Part 2 – Pavement Maintenance, Rehabilitation and Overlay Design	RDM 5.2
6	Road Design Manual: Vol. 6 Traffic Control Facilities & Communication Systems Design	Part 1 – Road Marking	RDM 6.1
		Part 2 – Traffic Signs	RDM 6.2
		Part 3 – Traffic Signals and Communication System	RDM 6.3
		Part 4 – Other Traffic Control Devices	RDM 6.4
7	Road Design Manual: Vol. 7 Road Lighting Design	Part 1 – Grid-connected Road Lighting	RDM 7.1
		Part 2 – Solar Road Lighting	RDM 7.2

This Manual must be applied sensibly along with the skill and judgement of the auditors. Compliance with the guidance given in the Manual does not relieve the auditors of the responsibility of exercising their knowledge and training in conducting audits.

1.2 Objective

The objective of this manual is to assist users to carry out road safety audits systematically. It guides auditors to take a proactive approach to assess possible safety hazards and recommend actions so that a road crash does not happen. The overall objective is to prevent road crashes by applying a safe system approach in design, construction, pre-opening and post-completion stages and existing roads.

1.3 Scope

The scope includes providing an overview of what road safety audit is, explaining how to conduct an audit in a step-by-step manner. An audit can be carried out at various stages of a road project:

- For feasibility / preliminary design stage, the Manual guides the auditors to look at the standards adopted, route choice, alignment, and cross sections.
- For the detailed engineering stage guidance is given on assessing the geometric layout, road marking, signage, intersections, and roadside clearance.
- For the construction stage, the auditors are guided to look at safety related to traffic management, work zone, pedestrian, and cyclist movement through the construction area.
- At pre-opening stage, the new road is ready to be opened to traffic. The Manual guides the auditors on what safety deficiencies to look for.
- For post-completion audit, the manual gives guidance on the possible safety deficiencies on a newly constructed road. This includes proper signage, marking, placement of guardrails, pedestrian facilities, and lighting.
- The post-completion guidance can also be used for audit of existing roads.
- For existing roads, the Manual guides the auditors to look at deficiencies at intersections, presence of roadside hazards, inadequate facilities for pedestrians, signage, and marking, so that safety can be improved.

The Manual describes how to include a safe system approach while auditing road projects. Technical tips have been provided to assist the auditors. A draft Terms of Reference has been included. Examples have been provided from various stages of audit to guide the auditors. Checklists have been provided to help the auditors in getting ready for conducting audits and to prepare audit reports.

1.4 Organisation of the Manual

The Manual is organised in 3 chapters and 5 appendices.

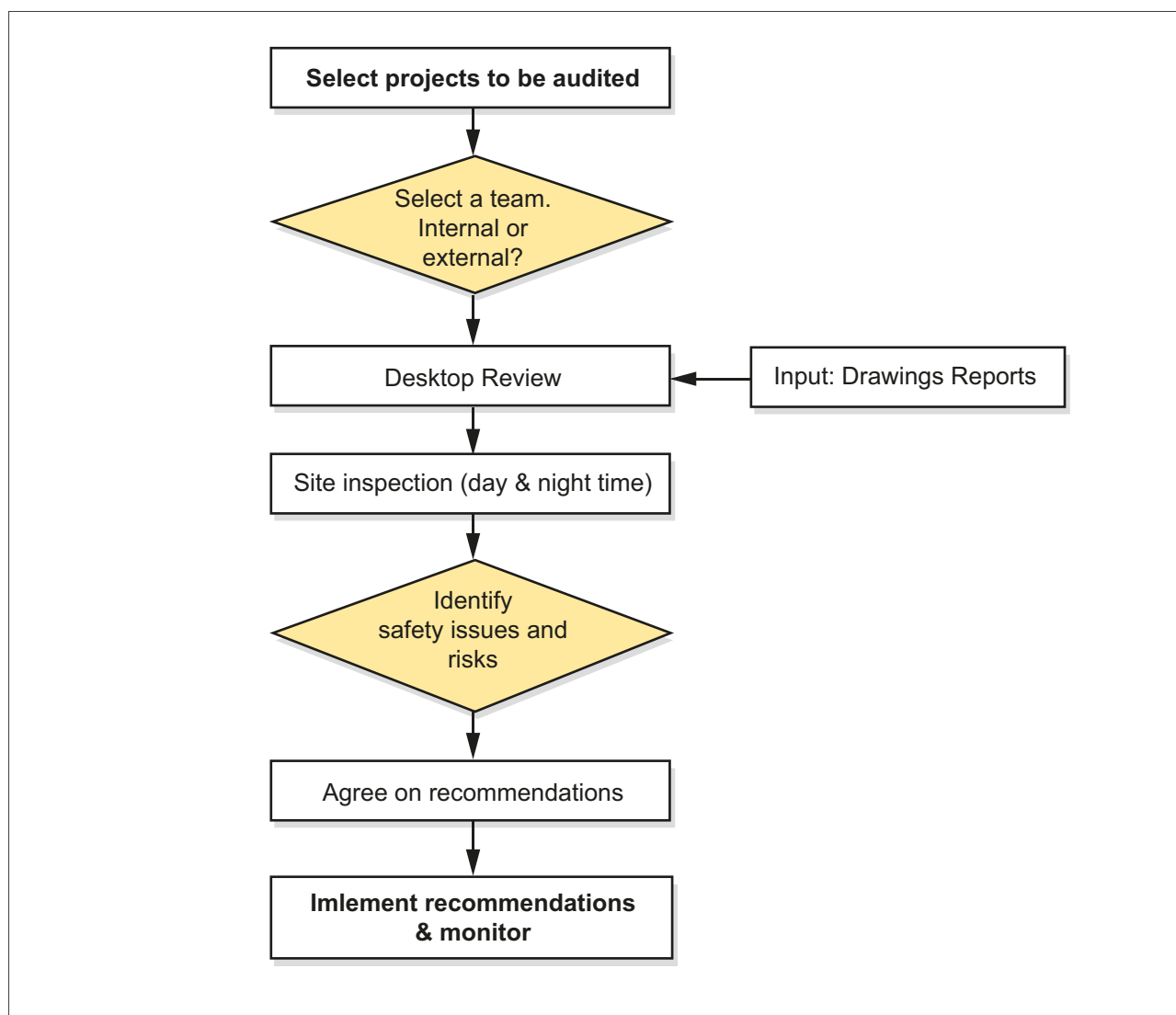
- **Chapter 1** details the objective, scope, organisation of the manual and the design process.
- **Chapter 2** provides an overview of road safety audit. It includes background, the need for audit, definition, stages of audit and responsibilities of key players.
- **Chapter 3** deals with conducting a road safety audit. It includes selection of a team, providing inputs, holding a commencement meeting, desk review and site inspection, writing reports, meeting to agree on the recommendations and follow up actions.
- **Appendix A** presents checklists for the different stages of an audit.
- **Appendix B** shows case studies from previous audits at different stages.
- **Appendix C** explains the various types of crash data that may be useful to an auditor.
- **Appendix D** explains the safe system approach and gives tips for auditing.
- **Appendix E** provides a sample Terms of Reference and audit brief contents.

This manual should be read in conjunction with the RDM Volume 1, Part 3.

1.5 Design Process

The process of carrying out a Road Safety Audit (RSA) is shown in Figure 1.1.

Figure 1.1 RSA Process



2 Overview

2.1. Background of Road Safety Audit

Road safety audits began in the late 1980s in England. A policy was developed requiring all new road designs to be checked and approved for safety by the Road Safety Engineering (RSE) team prior to construction. This checking process became formalised as the Road Safety Audit (RSA). The RSA team applied their skills in a proactive way to eliminate safety concerns at the design stage.

The road safety audit process has expanded globally since then. Many road agencies have adopted the RSA process and tailored it to suit their local needs.

2.2 The Need for Road Safety Audit

Road traffic crashes are a major issue in Kenya. The Economic Survey 2023¹ reported 10,210 traffic accidents in 2021 where 4,579 persons were killed, 10,050 persons were seriously injured and 5,996 persons were slightly injured. Pedestrians and motorcyclists are most at risk, with 1,558 pedestrians and 1,270 motorcyclists killed in 2021.

It is estimated that 13 people are killed and 44 are injured every day in Kenya, in road crashes, and 67% of fatalities and injuries are in the economically productive age groups of 15 - 64 years².

The World Bank Global Road Safety Facility estimated that the cost of fatalities and serious injuries in Kenya in 2016 was 6,314 million USD which was equivalent to approximately 9.1% of the country's GDP. There is a need for coordinated action by all the key stakeholders to address this serious concern. Road Safety Audits on road projects is one critical step in that direction.

2.3 Definition of a Road Safety Audit

An RSA is a formal safety examination of a road project by an independent and qualified audit team. It qualitatively estimates and reports on the potential road safety issues and identifies opportunities for improvements in safety for all road users.

An RSA follows a set process that leads to a formal report. Detailed safety examination requires time, knowledge, skill and judgment, with the audit team being independent from the design team. The audit team members are qualified and experienced in road safety engineering practices. Other professionals can add inputs to an audit, but the main auditors usually have an engineering background.

The outcome of an RSA is a report that identifies road safety issues and makes recommendations to mitigate against the negative impacts of these issues. Responsibility to implement these recommendations remains with the road authority.

An RSA assesses safety of all road users, not only of motor vehicle users. Safety of pedestrians, cyclists and other non-motorised users are assessed along with the safety of users of cars, motorcycles, buses and trucks.

Designers undertake the design of roads in accordance with the relevant design standards. Compliance with design standards, while important, does not necessarily result in an optimally safe road design. Auditors do not need to check compliance with standards. The auditors will put themselves in the shoes of the road users and anticipate the possible safety hazards in the design that may result in a crash.

¹ Economic Survey 2023, Kenya National Bureau of Statistics

² Road Safety Country Profile, Kenya, Global Road Safety Facility, The World Bank Group

What a Road Safety Audit IS

- ✓ Focused on road safety.
- ✓ A formal examination of the road project.
- ✓ Proactive in nature and tries to assess the possible safety risks.
- ✓ Conducted by an audit team that is independent of the design team.
- ✓ Conducted by persons with appropriate qualification, training and experience.
- ✓ Considers safety risks of all types of road users.
- ✓ Qualitative in nature.

What a Road Safety Audit is NOT

- ✓ A check of compliance with design standards.
- ✓ A substitute for regular design checks.
- ✓ A crash investigation or black spot analysis.
- ✓ A redesign of the project.
- ✓ A detailed site inspection.
- ✓ A way of assessing or rating a project as good or bad.
- ✓ A safety review.

There are other approaches for improving road safety, such as crash investigation, black spot analysis, road safety review, road safety inspection and International Road Assessment Programme.

Crash investigation is on-site investigation and collection of evidence to determine the events and circumstances which contributed to the occurrence of a crash and / or to the severity of the outcome. Past crash data on roads and intersections are studied and mitigation measures are suggested to improve the situation. Blackspots are the locations where multiple fatal and serious injury crashes have occurred over the past few years. Mitigation measures are suggested after studying the crash history. Both are reactive approaches to road safety management. On the other hand, RSA is a proactive approach, it attempts to reduce the risk of crashes through a systematic review and change of the design or by implementing countermeasures.

Road safety review is usually carried out by designers and/or safety experts. The reviewers do not have to be independent of the design team, whereas the primary requirement of a RSA is that the audit team are independent of the design team.

Road Safety Inspection (RSI) is an on-site systematic review of an existing road section to identify hazardous conditions, faults or deficiencies that may lead to road crashes. It focusses on finding and fixing the safety issues on existing roads, whereas RSA assess the risk of crashes and recommends measures for improvement for new schemes.

International Road Assessment Programme (iRAP) is a road safety management tool that can perform crash risk mapping, star rating for roads and can be used for safer road investment planning. RSA and iRAP can be complimentary.

2.4 Stages of a Road Safety Audit

Audits can be conducted at different stages of project development, i.e. planning, design, implementation, operation, and maintenance. (Refer to Roads Manual A: General – Procedures and Standards Manual for stages of project development.) Whether an audit needs to be performed at a particular stage depends on how important the stage is, if there is a need to conduct an audit at that stage and the prevalent practice. For the current situation in Kenya, an RSA can be conducted at five stages as described below.

1. Stage 1: Feasibility or Preliminary Design Stage

The Stage 1 audit can be for either feasibility or preliminary design stage. As there is no significant difference in safety aspects for feasibility and preliminary design, it is recommended that the two be considered in one stage. An audit on completion of the planning or feasibility study stage examines features such as design standards, route choice and continuity with the existing adjacent network, horizontal and vertical alignments, cross sections, and interchange/intersection layouts. Careful auditing at this early stage can help to reduce the costs and delays associated with changes that may otherwise be brought about during later audits.

2. Stage 2: Detailed Design Stage

The audit is carried out after completion of the detailed road design but before the preparation of contract documents. Typical considerations include geometric layout, pavement markings, signals, lighting, road signs, intersection details, clear sight triangle (visibility splay), clearances to roadside objects (crash barriers/frangibility, kiosks, advertisement boards) and provision for vulnerable road users. Attention to detail at the design stage can reduce the cost and disturbance associated with changes that may come up during a pre-opening audit.

3. Stage 3: Construction Stage

The audit takes place during construction of the road works. It examines the safety of the traffic management plans for each phase of construction for large road projects. It also inspects the provisions for road safety at the work site during the construction period. Typical issues examined include the provisions for pedestrian safety, advanced warning zones, adequate transition zone lengths, worker safety, effective numbers of reflective signs, safe delineation, credible speed limits, temporary crash barriers, lighting, and diversions.

4. Stage 4: Pre-Opening Stage

The audit involves a detailed inspection of the new road project prior to its opening. Although most road projects are constructed 'under traffic' there is a time just before the Contractor hands over the project when the project is almost complete. This is when a pre-opening stage audit is undertaken. The new road should be driven, ridden and walked (as appropriate) by the audit team to ensure that the safety needs of all road users are provided for. A night-time inspection is particularly important to check the installation and visibility of signs, markings, delineation, lighting and any other night-time/low light related issues.

5. Stage 5: Post-completion and Existing Road audit

Post-completion stage is when a new road has been opened to traffic for some time and an audit is done to determine if there are any safety deficiencies during operation. Often several safety deficiencies become apparent during operation, which may have not been identified in the earlier stages. Proper signage, marking, guidance to traffic, placement of guard rails, pedestrian walking and cycling areas, facilities for persons with disabilities, accessibility and lighting are emphasised. The project should be inspected at daytime and at night to observe the traffic operation. Auditors should walk on the stretches of roads that appear to have safety risks for pedestrians / cyclists / Persons with disabilities (PwD). This stage is also relevant for audit of existing roads.

The audit of existing roads aims to ensure that the safety features of a road are compatible with the functional classification of the road. It also aims to identify any feature that may develop over time into a safety issue (such as a tree blocking sight lines at an intersection or issues of visibility splay when sight triangles are obstructed).

A number of the safety issues typically found during audit can be readily addressed through simple and low-cost maintenance practices (e.g. tree trimming, replacing signs, placing road marking, and removal of roadside hazard issues).

The focus of the post-opening audit is to study whether all the safety features have been implemented in a proper manner and to observe the possible safety issues due to operations on the roadway.

2.5 Responsibilities of Key Players in Road Safety Audit

There are three key groups involved in the road safety audit process: the Client, the Designers, and the Audit Team.

The Client

The Client is the organisation or the 'owner' of the road on behalf of the government. The Engineer is the technical representative of the Client who looks after the day-to-day matters. The Client can also be a private investor (Concessionaire) for toll roads. The road authority is the final Client and takes the ultimate decisions.

Designers

This is the team commissioned by the Engineer (on behalf of the Client or by the Concessionaire) to design and develop the road project. The design team can be a part of the Client organisation or a consulting company.

Audit Team

The Audit Team usually has two or more persons who are qualified as road safety auditors and who are independent of the design team. The audit team is engaged by the Engineer on behalf of the Client. The audit team may also be from the Client's organisation, but they must be independent of the project.

The Audit Team Leader should be a civil engineer with sufficient experience in design, construction and maintenance of road projects. This person should be an experienced road safety auditor. The team members should have experience in design, construction, and maintenance with training in RSA.

The main functions of the key players in the road safety audit are summarised in [Table 2.1](#).

Table 2.1 Main Functions of the Key Players in Road Safety Audit

Key Players	Main Functions
Client / Project Owner	- Express a commitment to road safety. - Provide funding and resources for safe roads. - Consider safety audits as essential. - Commission audits at appropriate times. - Select road safety audit team. - Facilitate response to the recommendations of audits and arrange implementation of those recommendations that are accepted and agreed. - Attend commencement and completion meetings.
Design Team	- Attend commencement and completion meetings. - Provide relevant information to audit team. - Support the Client in responding to recommendations of audit. - Act on the audit recommendations.
Safety Audit Team	- Identify the safety issues in the proposed design / project. - Inspect the site during day and night. - Give constructive recommendations to reduce the risk of crashes and their severity. - Prepare report on safety concerns and recommendations. - Hold commencement and completion meetings with the Client and the design team.

3 Conducting a Road Safety Audit

3.1 Selecting the Road Safety Audit Team

A road safety audit is performed by a team comprising of members with expertise in road safety. Typical background of auditors includes traffic/transportation engineering, highway design and construction, crash investigation and analysis and human factors/road-user behaviour.

RSA requires qualification, experience and skill to assess the possible safety risks on a road project. Effective auditors must have experience in road safety engineering, an aptitude for accident investigation and prevention (AIP) techniques, together with experience in traffic engineering and traffic management, road design and road construction techniques.

The audit team needs to interpret technical drawings and design reports. The audit team should produce a clear and concise audit report so that the Engineer can understand the safety concerns detected. Otherwise, the Engineer may take decisions leading to waste of public money or to unsafe outcomes for the road users.

To engage a good audit team, the Engineer should state in the Terms of Reference that an audit is to be performed by a team (minimum of two persons, preferably three) that has the necessary experience and expertise in road safety engineering.

The team's competencies should cover:

1. Traffic engineering.
2. Road design and construction techniques.
3. Crash investigation techniques.
4. Road user behaviour.

For specific projects, expertise may be required relating to environmental and social aspects, law enforcement and right-of-way. The audit team leader can include such specialised person in the team, if required.

The Engineer should appoint the audit team and define the requirements of the audit in the Terms of Reference.

The RSA Team Leader should be a Senior Road Safety Auditor, with adequate qualification and experience in road safety engineering. The Team Leader is responsible for managing the audit, communicating with the audit team and ensuring that the report is completed on time. The Team Leader is usually the most experienced member of the audit team and provides technical guidance and leadership to the other team members.

In appointing the audit team, the Engineer should consider the following:

1. The auditor is independent of the design and implementation team.
2. The auditor has attended an approved audit training program and is a certified auditor.
3. The auditor has the necessary qualifications, experience and skills for this size and stage of project.

Independence of the audit team from the design and implementation team is very important. This is to ensure that the auditors can work impartially and identify the safety risks using professional judgement. The team members should not have prior experience of working on the same project at a different stage of design / construction, to prevent bias or a conflict of interest.

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The size of the RSA team depends on the size and nature of the project. Generally, a team of two to three members may be considered for most audits. Additional specialised experts may be included as required. For minor projects on low volume and low speed roads, the team may consist of one Senior Road Safety Auditor along with one Trainee Auditor who is a qualified civil engineer and trained in road safety.

3.2 Providing all the Background Information

The Engineer should provide an Audit Brief to the audit team which is specific to the project. The brief should define the scope and details of the road or development project to be audited, including sufficient information for the audit to be undertaken. The level of detail required depends on the stage of audit. A set of all relevant project documents is to be provided to the audit team.

The information provided should include:

1. Project description.
2. Design principles and standards that were used.
3. Description of any departures from approved standards and the reason for departure.
4. Traffic data and traffic forecasts.
5. Data on pedestrian movements.
6. Information on public transport services, including the location of stopping places.
7. Crash history.
8. Full set of drawings showing details of the horizontal and vertical alignment and other features.
9. Signage and marking plans (essential for Detailed Design audits)
10. Copy of the previous audit reports (if any) and an account of any changes since the previous audit (if any).

An example Audit Brief contents is given in [Appendix E](#).

3.3 Holding a Commencement Meeting

The project documents need to be handed over to the audit team by the designers during the commencement meeting. The project and the process of the audit can be explained at this time. The major points in relation to this are:

1. The auditor will make safety recommendations in a formal report.
2. Designers will propose changes to the design based on the recommendations, or formally reply as to why no changes are needed.
3. Client takes the final decision on design changes.
4. All design changes are made by the designer, not by the auditor.
5. Some preliminary discussions with the designers are beneficial to clarify areas of concern.
6. The auditor should maintain their independence and not get involved in redesigning elements of the project.
7. The Resident Engineer of the Project Supervision Team should attend the commencement meeting for Stage 3 (Construction Stage) and Stage 4 (Pre-Opening Stage) road safety audits.

3.4 Checking the Drawings, Design Reports and Documents

All drawings and documents should be reviewed before site visit and then continuously during the audit. The auditors review the plans, profiles, and other drawings to make preliminary assessments of items of interest or concern. This is done by marking the drawings and circling the items of interest. Figure 3.1 to Figure 3.3 are examples of a marked-up drawing highlighting audit concerns over a guard rail on a bridge deck and approaches.

Figure 3.1 : General Locations of Concern

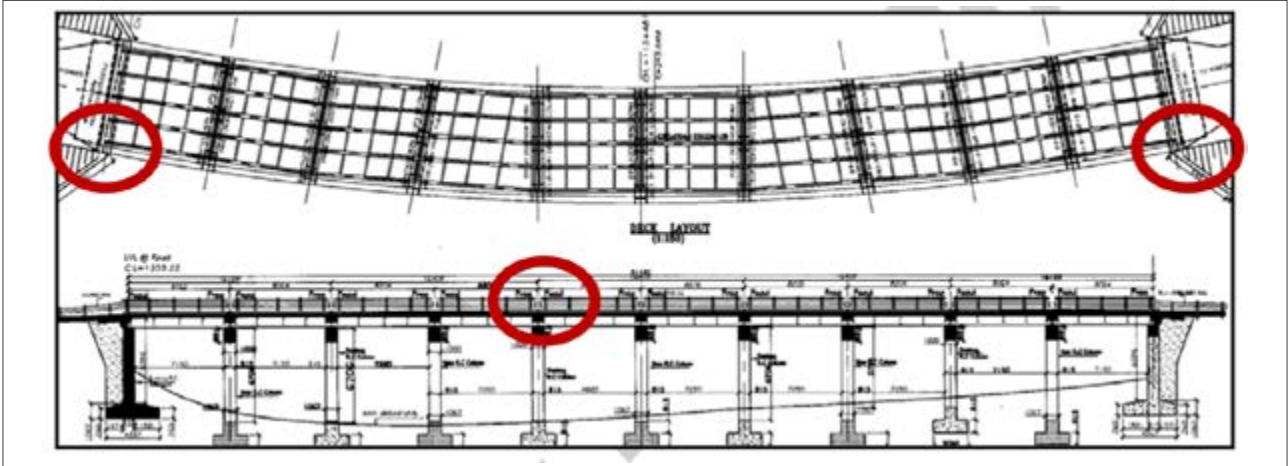


Figure 3.2 Locations of Concern Over Guard Rail Terminal and Gap in Parapet

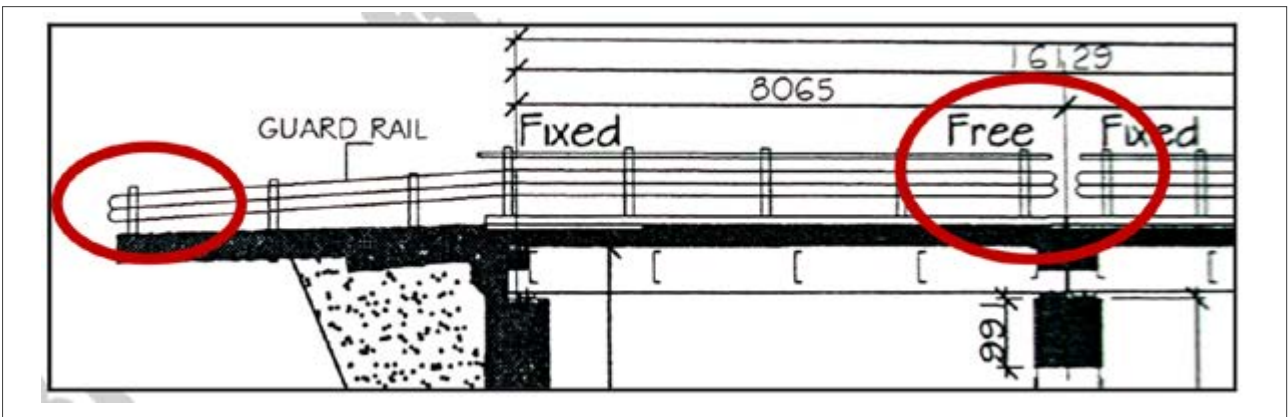
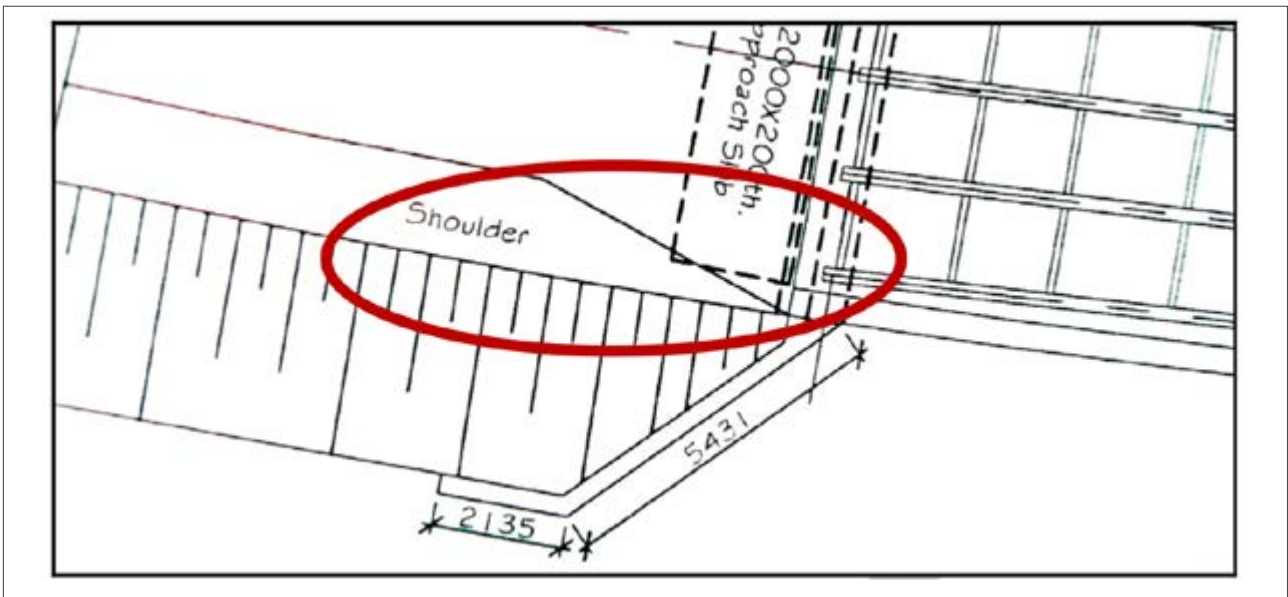


Figure 3.3 Lack of Detail of Guard Rail Terminal



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3.5 Inspecting the Site

The road proposal should be inspected by the entire audit team, during daytime and nighttime. This inspection involves carrying the drawings and reports to the site and comparing them with the site situations. It involves imagining what the finished road project will look like and how it will operate. The team members should look at the proposal from the viewpoint of all the road users.

Many photographs should be taken, some of which can be used in the audit report. The rest can be kept as a record of the audit assignment for future reference. A record should be kept of the observations either on paper or digitally.

While inspecting the site, auditors need to anticipate whether different light conditions (day and night) and weather (e.g. fog, rain) conditions may create safety concerns. Transition zones, where the new road merges with an existing road can pose risks and should be observed carefully.

3.6 Risk Assessment of Safety Concerns and Prioritisation of Suggested Recommendations

When there are several safety concerns, the Client needs assistance to assess the risk involved with each issue and to prioritise the recommendations that can be adopted.

Risk is defined as the product of the likelihood (frequency) of an event and the impact (severity) of that event if it occurs. Table 3.1 shows the relationship between likelihood and the severity of an event. If an event falls in the red zone, it is of very high or high risk and it must be addressed. If an event falls in the orange zone, it is of medium risk. The events in the yellow zone are of low risk.

Table 3.1 Main Functions of the Key Players in Road Safety Audit

Severity/Risk Level	Likelihood of Occurrence				
	Frequent	Probable	Occasional	Remote	Improbable
Catastrophic	Very high risk	Very high risk	High risk	Medium risk	Medium risk
Critical	Very high risk	High risk	High risk	Medium risk	Low risk
Major	High risk	High risk	Medium risk	Low risk	Low risk
Minor	High risk	Medium risk	Low risk	Low risk	Very low risk
Negligible	Medium risk	Low risk	Low risk	Very low risk	Very low risk

Table 3.2 provides simple criteria for assessing the risk associated with the safety concerns found by the auditor. It is a qualitative assessment and requires professional judgment and experience of the audit team.

Table 3.2 Criteria for Risk Assessment

Risk	Severity	Examples
Very High	Multiple deaths can occur.	- High speed, multi-vehicle collision on an expressway. - High speed bus collision with a bridge abutment.
High	A death and/or serious injuries can occur.	- High or medium speed vehicle to vehicle collisions. - High or medium speed vehicle collision with a fixed roadside object. - Vehicle pedestrian collision.
Medium	Minor injuries can occur.	Low speed collisions, such as a three-wheeler (tuk-tuk) collision with a cyclist, a rear end collision in a slip lane, a pedestrian struck in a car park.

Table 3.3 gives guidance on assigning a priority level to each recommendation. The priorities can be considered by the Engineer in view of the constraints in implementation.

Table 3.3 Assigning Priority Level for Suggested Recommendations

Priority	Suggested Treatment Approach
Essential	Where risk is assessed as Very High, the recommendation should be implemented as soon as possible.
Highly Desirable	Where risk is assessed as High, the recommendation should be implemented, unless the cost is prohibitive. If the cost is prohibitive the risk can instead be reduced by applying an alternative measure.
Desirable	Where risk is assessed as Medium, the recommendation may be implemented to reduce the risk.

3.7 Writing the Road Safety Audit Report

Audit reports should be concise and technically clear. They should follow an agreed and uniform format. Each safety concerns shall be clearly stated.

The audit report should contain:

1. A title page showing name and location of the project.
2. Brief description of the project and stage of audit.
3. The names of the road safety audit team members.
4. A list of all drawings, reports and documents reviewed as a part of the audit.
5. Dates of the site inspections and the weather conditions at that time.
6. Any available traffic volume and crash data.
7. A listing of all the safety concerns in a tabular form with photographs and chainage/ location.
8. A risk rating for each safety concern.
9. A practical and clear recommendation for corrective action for each safety concern.
10. A statement signed and dated by the Team Leader, indicating that the team has audited the drawings/documents, inspected the site, identified the road safety concerns and provided recommendations.
11. A Corrective Action Report (CAR), in a tabular form, which is to be completed by the Client to respond to the findings and recommendations.

3.8 Holding a Completion Meeting

The Engineer is responsible for arranging a completion meeting involving:

1. The Audit Team Leader.
2. The Engineer.
3. The Designer/Design team.

The meeting provides an opportunity for the auditor, the Engineer, and the design team to discuss all the issues in the report. This involves a discussion of each safety concern, its risk rating, priority, and ways to address the problem.

The meeting should be held in a professional and co-operative manner with all parties working together to enhance the safety of all road users. The meeting should not become about apportioning blame or a serious dispute.

The Team Leader must not entertain any suggestion of altering a report to soften the audit findings.

3.9 Providing Response to the Audit Report

The Engineer is required to respond in writing to each recommendation in the RSA Audit report.

The Engineer can:

1. Accept the report completely (and develop solutions to overcome or reduce the safety concern).
2. Accept the safety concern but not agree to the recommendations. In such cases, the response should set out alternative ways to resolve the safety concern.
3. Not accept the recommendation (explaining clearly why this decision has been taken).

To provide useful feedback, the Engineer should send a copy of the response to the RSA Team Leader for information. The audit team should note the responses and where possible, learn from them. The team should not create a dispute over which recommendations have or have not been accepted.

The audit team provides guidance, but it is the Engineer and the designers who are responsible for the final decisions about potential solutions/re-design.

3.10 Deciding the Way Forward

The Engineer should consider the time and cost implication of the recommendations given by the auditor. If the recommendations are not acceptable in part or full, the Engineer may ask the Designer to come up with an alternative solution. The alternative solution and the reason for it should be recorded.

3.11 Following up and Implementing the Agreed Recommendations and Changes

Safety on roads cannot be enhanced unless the agreed audit recommendations are implemented. After the Engineer is satisfied with the recommendation of the audit team or the alternative solution prepared by the design team, the solutions should be implemented in the design or on the ground.

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Appendix A – Road Safety Audit Checklists

A.1 Purpose of Checklist

Checklists have been developed for use in road safety audits. The checklists are to assist the audit team to consider the safety needs of the vulnerable road users (pedestrians, bicyclists, NMT, motor cyclists) and motorised road users (car, truck, and bus users).

Checklists are intended to guide the auditor in identifying risk so that some risks do not get overlooked during an audit. However, checklists are not a substitute for knowledge or experience. Every project is different, based on the type of road, topography, traffic, user behaviour. Judgement should be used while referring to the checklists. It can be expanded or modified as per the requirements of the project. Filled in checklists do not constitute an audit report. Checklists help in writing a report.

Checklists are given for five stages of audit.

The checklists have been worded such that a negative answer (NO) to any question means there is a safety issue.

A positive (YES) answer to any question means that issue has been examined, it is not considered as a safety issue.

The NA (Not Applicable) column has been provided for cases where the issue may not be relevant.

The comments column should be used for writing specific comments on the issue.

Table A.1 Checklist for Stage 1: Feasibility / Preliminary Design

Checklist for Stage 1: Feasibility/Preliminary Design					
Sl. No.	Issues	Yes	No	NA	Comments
A	General topics				
1	Is the project consistent with the development plans for the area?				
2	Is the project consistent with the planned road hierarchy for the area?				
3	Will the project cater for all modes of traffic? <i>(cars, buses, goods vehicle, pedestrians, cyclists, motorcyclists, matatu, NMT)</i>				
4	Does the project serve major generators of traffic safely? <i>(for existing or planned facilities like offices, factories, etc.)</i>				
B	Road Design Standard				
1	Are the design standards being used appropriate <i>(considering the class of terrain, mix of traffic including Vulnerable Road Users (VRU) and the function of the proposed road)</i> ?				
2	Has the design speed been adopted considering the hierarchy?				
3	Will the likely speed limit on the proposed road be compatible with the design speed and mix of traffic, including non-motorised traffic and pedestrians?				
4	Does the route fit in with the physical constraints imposed by the topography?				
5	Does the project road relieve routes or sites with bad crash records?				
6	Are the alignment standards consistent all along the route?				

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Appendix A – Road Safety Audit Checklists

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Appendix A – Road Safety Audit Checklists

Checklist for Stage 1: Feasibility/Preliminary Design

Sl. No.	Issues	Yes	No	NA	Comments
C	Alignment				
1	Will the design speed be 'safe' with regard to horizontal and vertical alignment?				
2	Does the alignment (<i>horizontal and vertical</i>) give sufficient forward visibility for the selected design speed? (<i>Check for inadequate stopping sight distances</i>)				
3	Will horizontal and vertical alignments be safe and consistent, especially at interchanges and intersections?				
4	Will all merging, diverging and weaving areas be 'safe'?				
5	Do the horizontal and vertical alignments fit together comfortably? (<i>Check for bad combinations, such as a sharp bend immediately after a summit curve, and sag curve within a bend</i>).				
6	Will sight distances be satisfactory – especially at intersections and property accesses?				
7	Does the alignment provide safe overtaking opportunities?				
8	Does the treatment at curves have adequate and safe provision for transition curves, super-elevation and extra width of carriageway and formation?				
9	Has the vertical alignment been designed such that maximum grades are interspersed with recovery grades?				
D	Visibility and Sight Distance				
1	Are sight and stopping distances adequate throughout the project length?				
2	Is the design free of sight restrictions (<i>due to buildings, trees, signs or rock slopes</i>)?				
3	Does the design provide adequate visibility for an approaching driver to see a pedestrian waiting to cross the road?				
4	Does the design provide an adequate visibility funnel for an approaching driver to see a vehicle waiting to cross the carriageway from a side road or storage lane of the median?				
E	Staging of Work				
1	Are the stages of construction arranged to ensure safety during construction?				
2	If the construction stage is for road widening or dual carriageway, is the design adequate to give the message to the driver?				
3	Has the transition between single and dual carriageway been designed in a safe manner?				

Checklist for Stage 1: Feasibility/Preliminary Design

Sl. No.	Issues	Yes	No	NA	Comments
F	Cross Section				
1	Are the widths of the carriageway, shoulders, medians, and service roads in accordance with standards and adequate for the function of the road and the volume and the mix of traffic likely to use it?				
2	Do bridges/culverts have walkways with proper gradients/crash barriers/parapets?				
	• On existing bridges / culverts				
	• On proposed bridges / culverts				
	• On approaches to vehicular or pedestrian underpasses				
3	Is the cross section consistent along the project length? <i>Note: any inconsistency</i>				
4	Has the shoulder been designed to a safe standard? <i>Note: any inadequate width of shoulder</i>				
5	Have the side slopes been designed to a safe standard?				
6	Are the side slopes safe for any run-off vehicles to traverse?				
7	Does the cross section have adequate provision for VRU?				
	• Pedestrians: Have paved walkway, adequate refuge width on median and accessibility ramps been provided?				
	• Cyclists: Have segregated areas (<i>e.g. paved shoulders</i>) been provided where numbers are significant.				
8	Is the cross section wide enough to provide sheltered turn lanes within the median?				
9	Is it possible to provide a storage lane for small vehicles that need protection from exposure to through traffic?				
10	Is the median wide enough to install street lighting?				
11	Does the cross section avoid unsafe compromises at bridges and other narrow areas? <i>Note: For safety, it is desirable to continue full width shoulders across bridges and other narrow areas.</i>				
12	Are overtaking lanes provided?				
13	Are climbing lanes provided?				
14	Have descending lanes and escape ramps been provided?				
15	Have all roadside hazards been managed safely? <i>Note: Check if adequate clear zone is provided</i>				

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Appendix A – Road Safety Audit Checklists

Checklist for Stage 1: Feasibility/Preliminary Design

Sl. No.	Issues	Yes	No	NA	Comments
G	Intersections				
1	Is the proposed type of junction (<i>T, staggered, roundabout, signal controlled</i>) suitable for the functional class of the intersection roads?				
2	Is the proposed junction suitable for the traffic volume and movements (<i>vehicles, pedestrians, NMT</i>)?				
3	Is the proposed type of junction the safest alternative?				
4	Does the layout encourage slow, controlled speed at the approach to stop/give way lines and other critical decision points?				
5	Are the number of junctions proposed and their distribution suited to the function of the road considering the surrounding network and access needed?				
6	Are the median openings for U-turns proposed at locations to minimise wrong way movements?				
7	Are the median openings clearly visible to the approaching traffic?				
8	Would the junction be visible and easily understood by vehicles from all approaches?				
9	Has channelisation been provided for different streams of traffic?				
10	Have right turn lanes, acceleration and deceleration lanes been provided?				
11	Is it safe for pedestrians and NMT to use the junction?				
12	Can all the turns be made safely?				
13	Are sight lines at each intersection adequate and free of obstruction (<i>buildings, structures, traffic signs, vegetation, and encroachment</i>)?				
14	Has a Y junction been avoided?				
15	Is the proposed traffic control at each intersection (<i>Stop/Give Way, roundabout, traffic signals</i>) as 'safe' as practical?				
16	Is the provision of night-time lighting adequate?				
H	Interchanges				
1	Is the type of interchange proposed suitable for the class of road?				
2	Are all features of the interchange designed in a safe manner?				
3	Is the spacing between interchanges adequate?				
4	Will the type of interchange be understood by road users?				

Checklist for Stage 1: Feasibility/Preliminary Design

Sl. No.	Issues	Yes	No	NA	Comments
J	VRU (pedestrians, cyclists, motor cyclists, NMT)				
1	Will pedestrians (<i>children, women, old and disabled</i>) be able to safely walk along both sides of the road? <i>Note: Consider the expected traffic speed on the new road.</i>				
2	Will pedestrians (<i>children, women, old and disabled</i>) be able to cross the road safely? <i>Note: Consider the expected traffic speed on the new road.</i>				
3	Do VRU have connectivity along their route?				
4	Is the design free of 'squeeze' points where VRU may be exposed to traffic?				
5	Does the design provide an adequate width of walkway and ensure sight distance?				
K	Matching with existing road				
1	Have the interfaces between the existing and proposed roads been designed in a safe manner for all road users?				
2	Is the interface safe from hazards? (<i>sharp horizontal / vertical curve, road side objects, drains</i>)				
3	If the interface cuts across village roads, pedestrian paths, etc., have safe alternative routes been provided?				
L	General road safety issues				
1	Have safety measures been considered for special events like market days, festivals or other intermittent activities?				
2	Have railway level crossings been identified and safe treatments proposed?				
3	Are there sufficient truck rest areas proposed with toilet facilities?				
4	Are sufficient rest areas proposed?				
5	Will the new road be as safe as practical given the local weather conditions (<i>sunrise, sunset, fog, snow, dust storms, rain, storms</i>)?				
6	Is the proposal likely to be free of 'visual deceit'?				
7	Has the proposal taken safe account of animals on the road, including livestock and wild animals crossing from adjacent forests or fields?				
M	Roadside hazards				
1	Are there any roadside hazards which appear to have been left out in the design?				
2	Are crash barriers proposed only where necessary?				
3	Is the type of barrier proposed suitable for this location?				
4	Are impact attenuators provided in gore areas?				
5	Do the standard drawings indicate that the barrier terminals will be of a suitably safe type?				
6	Do the standard drawings indicate a safe connection of the barrier to bridge abutments, together with the necessary reduction in post spacing to stiffen the barrier and prevent 'pocketing'?				

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Appendix A – Road Safety Audit Checklists

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Checklist for Stage 1: Feasibility/Preliminary Design

Sl. No.	Issues	Yes	No	NA	Comments
7	Has the proposal taken safe account of animals on the road, including livestock and wild animals crossing from adjacent forests or fields?				
N	Provision for public transport				
1	Does the design provide for all forms of public transport that will use the new road?				
2	Are bus lay-bys to be provided at places where passengers are most likely to use them?				
3	Are walkways proposed to and from bus stops and other key public transport locations to assist passengers in safe commuting?				
4	Have suitable paved waiting areas been proposed for NMT?				
O	Night visibility				
1	Will the interface between the new road and the old road be obvious and as safe as practical at night?				
2	Will the proposed geometry be understandable for road users to traverse at night?				
3	Are electric poles safely located?				
P	Maintenance				
1	Will road maintenance vehicles be able to work safely on the new road?				
Q	Drainage				
1	Will the road be well drained?				
2	Will the drains be covered, or located behind the crash barrier?				

Table A.2 Checklist for Stage 2: Detailed Design

Checklist for Stage 2: Detailed Design					
Sl. No.	Issues	Yes	No	NA	Comments
A	General topics				
1	Has a road safety audit been carried out earlier?				
2	Are circumstances since the last audit the same? (traffic volume, mix, new development)				
3	Has the general form of the project design remained unchanged?				
4	Is the road network the same as for previous audit?				
5	Is the proposed function of the road still the same as intended?				
B	Horizontal and vertical alignment				
1	Is the design speed safe with regard to horizontal and vertical alignment?				
2	Does the horizontal and vertical alignment give sufficient forward visibility for the selected design speed? <i>Note: Are adequate stopping sight distances provided?</i>				
3	Will horizontal and vertical alignments be safe and consistent, especially at interchanges and intersections?				
4	Will all merging, diverging and weaving areas be safe?				
5	Is the proposal consistent with the adjacent road network?				
6	Do the horizontal and vertical alignments fit together comfortably? <i>Note: Check for bad combinations, such as a sharp bend immediately after a summit curve, and sag curve within a bend.</i>				
7	Will sight distances be satisfactory – especially at intersections and property accesses?				
8	Does the alignment provide safe overtaking opportunities?				
9	Does the alignment avoid creating situations where the forward visibility is marginal for overtaking vehicles?				
10	Have adequate and safe provision for transition curves, super-elevation and extra width of carriageway and formation been provided?				
11	Has the vertical alignment been designed so that the maximum grades are interspersed with recovery grades?				
C	Cross section				
1	Are the widths of the carriageway, shoulders, medians, and service roads in accordance with standards and adequate for the function of the road and the volume and mix of traffic likely to use it?				

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Appendix A – Road Safety Audit Checklists

Checklist for Stage 2: Detailed Design

Sl. No.	Issues	Yes	No	NA	Comments
2	Do bridges and culverts have walkways with appropriate gradient / crash barrier / parapets?				
	• On existing bridges / culverts				
	• On proposed bridges / culverts				
	• On approaches to proposed vehicular and pedestrian underpasses <i>Note: any location where the cross-section standard changes abruptly along the route or is otherwise inconsistent with driver expectations.</i>				
3	Have the shoulders and side slopes been designed to a safe standard?				
4	Are shoulder widths adequate?				
5	Have the side drains been designed to a safe standard?				
6	Does the cross-section have adequate provision for VRU including persons with disabilities?				
	• Pedestrians: is there provision for walkway, refuge island at medians, accessibility ramps?				
	• Cyclists: is there a segregated area where there is a significant number of cyclists?				
7	Is the cross section wide enough to provide sheltered turn lanes within the median?				
8	Is there provision of a storage lane for small vehicles that need protection from exposure to through traffic?				
9	Is the median wide enough for the safe installation of street lighting?				
10	Are all roadside hazards managed 'safely'? <i>Note: Is the clear zone adequate?</i>				
11	Are shoulders proposed to be paved?				
12	Are overtaking and/or climbing lanes proposed, especially in hill sections?				
13	Is safe provision to be made for breakdown vehicles and emergency vehicles?				
14	Have descending lanes and escape ramps been provided?				
D	Matching with existing road				
1	Have the interfaces between the existing and proposed roads been designed in a safe manner for all road users?				
E	Staging of Work				
1	Are the stages of construction arranged to ensure safety during construction?				
2	Has the transition between single and dual carriageway been designed in a safe manner?				

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Appendix A – Road Safety Audit Checklists

Checklist for Stage 2: Detailed Design

Sl. No.	Issues	Yes	No	NA	Comments
F	Intersections				
1	Is the proposed type of junction (<i>T, staggered, roundabout, signal controlled</i>) suitable for the traffic volume and movements (<i>vehicles, pedestrians, NMT</i>)?				
2	Does the layout encourage slow, controlled speed at the approach to stop/give way lines and other critical decision points?				
3	Are the number of junctions proposed and their distribution suited to the function of the road considering the surrounding network and access needed?				
4	Are the median openings for U-turns proposed to minimise wrong way movements?				
5	Can the median openings be clearly seen by the drivers of the approaching vehicles?				
6	Would the junction be visible and easily understood by vehicles from all approaches?				
7	Has channelisation been provided for different streams of traffic?				
8	Have adequate facilities been provided for pedestrians and NMT to use the junction?				
9	Are all pedestrian crossings marked and pedestrians channelised to the crossings?				
10	Can all the turns be made safely?				
11	Are sight lines at each intersection adequate and free of obstruction (<i>buildings, structures, traffic signs, vegetation, encroachment</i>)?				
12	Has a Y junction been avoided?				
13	Is the proposed traffic control at each intersection (<i>Stop/Give Way, roundabout, traffic signals</i>) as 'safe' as practical?				
14	Will traffic signal control at the intersection(s) be clearly seen and understood?				
15	With roundabouts, is adequate deflection provided for all approaches?				
16	Are correct signs shown for all approaches to the roundabout?				
17	For other intersections, are Stop/Give Way signs shown and correctly located for maximum visibility?				
18	Is there an adequate visibility funnel for an approaching driver to see a pedestrian waiting to cross the road?				
19	Is there an adequate visibility funnel for an approaching driver to see a vehicle waiting to cross the carriageway from a side road or storage lane of the median?				
20	Is the provision of night-time lighting adequate?				

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Appendix A – Road Safety Audit Checklists

Checklist for Stage 2: Detailed Design

Sl. No.	Issues	Yes	No	NA	Comments
G	Junctions with traffic signal				
1	Can the signals be clearly seen on the approach to the junction?				
2	Have speed reduction measures been taken for the approaches?				
3	Has the potential for confusion when groups of signals are placed close together (<i>see-through effect</i>) has been eliminated?				
4	Have signal hoods been fitted to prevent drivers from seeing signals that do not apply to them?				
5	Are signals placed such that they can be seen in bright sunshine?				
6	Are the signal heads fitted with backing boards?				
7	Are the signal lamps the correct size?				
8	Are there at least two signal heads (<i>primary and secondary</i>) controlling each traffic movement?				
9	For two or more lanes on the approach, has a second primary signal been provided?				
10	Are all directional movements shown in the signal clear?				
11	Is there sufficient lateral clearance between signal heads and the carriageway?				
12	Do the signal colours, arrangement, signal sequence, and signal timings conform to accepted practice?				
13	Have unexpected conflict situations due to signal phasing been eliminated?				
14	Have protected right turns been provided where necessary?				
15	Is the 'inter-green time' between conflicting phases sufficient for safe operation?				
16	Can the junction be used safely if the signals are not working or are switched to flashing amber?				
17	Is there a phase to accommodate pedestrians?				
18	Are the settings and timings adequate for safe use?				
19	Have pedestrian signals been properly placed to avoid confusion?				
20	Is the control equipment located in a safe place where it will not interfere with visibility and is unlikely to be hit by errant vehicles?				
21	Is there safe parking for maintenance vehicles?				
H	Roundabouts				
1	Is the geometry simple and easily understood by drivers on the approach to the roundabout?				
2	Is the size of the roundabout sufficient for the volume and mix of traffic and the number of entries?				
3	Is the central island sufficiently conspicuous?				

Checklist for Stage 2: Detailed Design

Sl. No.	Issues	Yes	No	NA	Comments
4	Are the number of entries suitable for safe efficient operation?				
5	Are the entries and exits spaced far enough apart?				
6	Does the design deflect entering traffic sufficiently to ensure that entry speeds are no greater than 50 km/h?				
7	Is the visibility for entering traffic and circulating traffic adequate?				
8	Has the centre island been designed to be forgiving to errant vehicles?				
9	Has adequate provision been made for pedestrians to cross the arms of the roundabout?				
10	Have the needs of cyclists and NMT been considered?				
11	Is the signage and marking adequate?				
I	Interchanges				
1	Are sight lines open and free of obstruction at all merges and diverges?				
2	Are the distances between decision making points sufficient for safety?				
3	Is the signage project for each interchange clear and will it be easily understood by road users?				
4	Are all roadside hazards in the interchange identified and safely protected?				
J	Bridges				
1	Is continuity of carriageway, shoulder, pedestrian walkway and cyclepath maintained on the bridge?				
2	Can the parapet act as a safety barrier?				
	Are the height, level of containment, and size of opening of the parapet as per safety requirements?				
3	Are the parapet ends properly protected?				
K	Safety barrier				
1	Are safety barriers provided where necessary?				
2	Are the barriers long enough to prevent an out-of-control vehicles from reaching the hazard?				
3	Are the terminal arrangements (<i>upstream and downstream ends</i>) safe?				
4	Is the positioning of safety barriers suitable so as not to restrict the visibility of users?				
5	Do safety barriers block pedestrian desire lines?				
6	Has the metal beam guardrail been designed correctly? <i>Note: check beam height, post spacing, lateral clearance, spacer blocks, nuts and bolts, reflectors, terminal pieces, anchorage and whether the beams are overlapped correctly.</i>				
	• Check that the transitions between barrier types (e.g. <i>steel beam guardrail to concrete bridge parapet</i>) are safe.				
	• Are all features of the safety barrier acceptable?				

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Checklist for Stage 2: Detailed Design

Sl. No.	Issues	Yes	No	NA	Comments
L	Access control				
1	Will all accesses points be 'safe'?				
2	Is fencing provided in rural areas to keep animals from straying onto the road?				
3	Have measures been taken to ensure the safety of access to schools, public areas and commercial centres?				
M	Roadside hazards				
1	Are crash barriers proposed only where necessary?				
2	Is the type of barrier proposed suitable for this location?				
3	Are impact attenuators provided in gore areas?				
4	Do the standard drawings indicate that the barrier terminals will be a suitably safe type?				
5	Do the standard drawings indicate a safe connection of the barrier to bridge abutments, together with the necessary reduction in post spacing to stiffen the barrier and prevent 'pocketing'?				
N	VRU (pedestrians, cyclists, motor cyclists, NMT)				
1	Do all VRU have connectivity along their route, with suitable lateral clearance to motor traffic?				
2	Will pedestrians (<i>children, women, old and disabled</i>) be able to safely walk along both sides of the road?				
3	Will pedestrians (<i>children, women, old and disabled</i>) be able to cross the road safely?				
4	If formal crossings are proposed, are these conspicuous on each approach?				
5	Are dropped kerbs provided at all intersections and mid-block locations where pedestrians are to cross?				
6	Are the correct signs and pavement markings proposed for each pedestrian facility?				
7	Will each crossing facility be well illuminated at night so that pedestrians can be clearly seen by drivers?				
8	If mid-block traffic signals are proposed, will these have pedestrian push buttons?				
9	Are pedestrian paths to be provided through medians to permit pedestrians to cross 'at road level' and to assist disabled pedestrians with improved access?				
10	Is the design free of 'squeeze' points where VRU may be exposed to traffic?				
11	Does the design provide an adequate width of walkway?				

Checklist for Stage 2: Detailed Design

Sl. No.	Issues	Yes	No	NA	Comments
0	Signage, Marking & Delineation				
1	Do all the signs (<i>regulatory, warning and direction</i>) shown in the drawings conform to the 6 Cs (<i>Conspicuous, Clear, Comprehensible, Credible, Consistent, and Correct</i>)?				
2	Are Information signs provided for the category/function of road proposed?				
3	Are overhead signs of appropriate size with adequate information and in prescribed languages?				
4	Is good delineation (<i>curve warning signs, advisory speed signs, guideposts and chevron alignment markers</i>) provided where required?				
5	Safety Audit of Variable Message Signs (<i>VMS</i>)				
	• Does the vertical signpost have proper horizontal clearance from the carriageway edge?				
	• Do the VMS signs have sufficient clearance at the highest point on the carriageway?				
	• Is the VMS sign clearly legible at day and night-time?				
6	Are the proposed signs of standard shape, size and colour?				
7	Are the proposed sign locations appropriate and at the right spacing?				
8	Are all junctions provided with an advance direction sign, distance information sign and intersection sign?				
9	Is the correct type of pavement marking proposed for different situations? <i>Note: lane line, edge line, firm and dotted line in the right places.</i>				
10	Is the correct colour of marking proposed? <i>Note: white, yellow, etc.</i>				
11	Is delineation of merging and diverging areas properly shown, including situations where 'through' traffic may inadvertently lead into auxiliary and turn lanes?				
12	Is it clearly demarcated which lanes the 'through' traffic needs to stay in?				
13	Are pedestrian crossing markings provided at junctions and mid-blocks of the sections?				
14	Is positioning of stop lines appropriate?				
15	Are the directional arrows marked on the pavement guiding the driver?				
16	Are 'Hazard markings' proposed at the approach ends of islands, medians, and culverts/bridges etc.?				
17	Have retro-reflective markers/road studs been proposed?				

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Checklist for Stage 2: Detailed Design

Sl. No.	Issues	Yes	No	NA	Comments
18	Has the correct colour of markers been proposed?				
19	Have correct chevron alignment markers been proposed?				
20	Do the drawings indicate high quality reflective material (<i>red on left, white on right</i>) at the top of each guidepost?				
P	Parking				
1	Have sealed and marked areas been provided for parking?				
2	Will the parking be adequate and safe?				
3	Have proper directional signs been provided for on-street / off-street parking areas?				
4	Have the beginning and end locations of parking been shown?				
Q	Emergency Service Vehicle Access				
1	Has provision been made for safe access and movements by emergency vehicles?				
2	Are median openings frequent, visible, and well signed?				
R	Lighting and night visibility				
1	Are all the signs visible at night?				
2	Are the critical locations (<i>intersections, pedestrian facilities, bus bays, bus stops, truck lay-by, toll plaza etc.</i>) proposed to be lit?				
3	Is lighting provided on road sections passing through built-up areas, service roads, above and below the grade separator, underpass, etc.?				
4	Are the proposed lighting project and illumination level as per appropriate standards, consistent with the needs of the location, pedestrians, and other factors?				
5	Will the proposed geometry be understandable for road users at night?				
6	Do the standard drawings indicate that the lighting columns will be frangible?				
7	If the columns are not frangible, have other actions been taken to make these safe for road users?				
8	Are all lighting poles in the median (<i>less than 2 m wide</i>) protected by crash barriers?				
9	Are all light poles placed at a suitable distance or protected by a crash barrier?				
10	Are all light poles clear of traffic signals or signs?				
S	Drainage				
1	Is sufficient drainage facility provided?				
2	Are the drains covered or located outside the clear zone or shielded behind a crash barrier?				

Checklist for Stage 2: Detailed Design

Sl. No.	Issues	Yes	No	NA	Comments
T	Utilities				
1	Is there adequate clearance for overhead power lines?				
2	Can utility apparatus be accessed safely?				
3	Can maintenance vehicles be parked safely?				
4	Are power boxes and access chambers located in a safe place (e.g. away from traffic lanes)?				
U	Vegetation and landscaping				
1	Have vegetation/ trees/ landscaping been located where they do not interfere with visibility and affect safety?				
V	Railway crossing				
1	Is the crossing located where visibility is adequate?				
2	Will there be adequate visibility to queue tails?				
3	Have necessary speed management measures on the approaches been considered?				
4	Have barriers and signals been designed in a safe manner?				
5	Have the Kenyan Railways authorities given approval for the crossing?				
W	Maintenance				
1	Can access to structures be carried out safely?				
2	Can maintenance vehicles stop in a safe place?				
3	Is planned maintenance adequate to deal with safety features such as shoulders and guardrails?				
X	Publicity				
1	Has the necessary road safety awareness campaign been conducted amongst roadside communities?				
Y	Enforcement				
1	Is planned policing adequate for the safety of the new road?				
Z	General road safety issues				
1	Is the new road as safe as practical for the local weather (<i>sunrise, sunset, snow, fog, rain, wind, storm</i>)?				
2	Is the project free of all 'visual deceit'?				
3	Is the road surface free of gravel and sand, and with good skid resistance?				
4	Has the proposal taken safe account of animals on the road, including livestock and wild animals crossing from adjacent forests or fields?				

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Table A.3 Checklist for Stage 3: Construction Stage

Checklist for Stage 3: Construction Stage					
Sl. No.	Issues	Yes	No	NA	Comments
A	Traffic Management Plan				
1	Has a traffic management plan (<i>TMP</i>) been prepared for the road work site?				
2	Has the TMP been approved by the Supervision Consultant or other authorised person?				
3	Does the Contractor have an appointed Safety Engineer who is responsible for road safety at the work site, including checking the condition of the installed TMP every day?				
4	Does the TMP provide adequate and correct signage, and delineation for all road users under all traffic conditions?				
5	Does the TMP provide work site protection (barriers) for all road users under all traffic conditions?				
6	Has the TMP been implemented correctly at the road work site?				
7	Is the advanced warning zone adequately signed to alert approaching road users of the presence of the road works?				
8	Is the transition zone correctly and adequately signed and delineated to guide approaching road users into their correct path?				
9	Is the termination zone adequately signed to advise road users that they are past the road works and may return to normal road/highway speeds?				
10	Are the flagmen highly conspicuous and placed where they can give clear instructions to approaching drivers/riders in advance of the work site?				
11	Has a safety buffer zone been provided?				
B	Speed control				
1	Are the speed limit signs conspicuous?				
2	Are there sufficient numbers of repeater speed restriction signs throughout the length of the work site?				
3	Is the signed speed appropriate for safe traffic movement through the work site?				
4	Have local Police been requested to enforce the signed speed limit through the worksite?				
5	Are speeds managed (<i>through signs, enforcement and if necessary, road humps</i>) so that vehicle operating speeds that pass within one traffic lane width of any work site are 40 km/h or lower?				

Checklist for Stage 3: Construction Stage

Sl. No.	Issues	Yes	No	NA	Comments
C	Signs, Signals, Pavement Marking and Delineation				
1	Are all necessary warning, direction and regulatory signs in place as shown in the TMP?				
2	Are all existing road signs (<i>those that are unnecessary during the works</i>) covered to avoid distraction or misinformation?				
3	Are all 'Men At Work' signs removed or covered when work is finished for the day? <i>Note: Applicable in case of maintenance works.</i>				
4	Do all signs satisfy the 6Cs of good signage practice? (<i>Conspicuous, Clear, Comprehensible, Credible, Consistent, and Correct</i>)				
5	Are temporary traffic signals clearly visible to the approaching drivers/riders?				
6	If the signals generate traffic queues, can the end of the queue be easily seen by approaching drivers/riders?				
7	Are pavement markings consistent and clear through the length of the work site for both day and night?				
8	Is delineation through the site safe for all road users under all anticipated conditions?				
9	With several layers of asphalt typically to be laid for new roads, is temporary lane marking being used on each new layer to guide drivers/riders as an interim safety initiative, especially at night?				
D	Diversions from one carriageway to another				
1	If traffic is to be diverted from one carriageway onto the other, do the advanced warning signs provide clear guidance about the diversion ahead and are they well located to alert all approaching road users?				
2	Is the transition zone well delineated (<i>with plastic cones and other forgiving devices</i>) to reduce the number of traffic lanes well before the carriageway ends?				
3	Is the carriageway (<i>on which the work is to take place</i>) fully closed at all intersections?				
4	Are drivers/riders from the side roads given adequate warning of the two-way traffic operation on the carriageway open on the main road?				
5	Is the carriageway (<i>on which the work is to take place</i>) fully and clearly closed off to all traffic?				
6	Are all drivers/riders in both directions on the other carriageway adequately reminded that they are on a two-way section of road (<i>such as with 'Two Way' warning signs</i>)?				
7	Is the management of all two-way traffic sections through the road work site safe for all road users both day and night?				

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Appendix A – Road Safety Audit Checklists

Checklist for Stage 3: Construction Stage

Sl. No.	Issues	Yes	No	NA	Comments
E	Road Surface				
1	Is the road surface suitable for safe movement by all road users, especially cycles and motorcycles?				
2	Are paved surfaces swept and kept free of gravel and sand?				
3	Has a suitable paved surface been provided along with correct lane markings to guide drivers/riders through diversions?				
4	Have signs and markers/road studs been used to highlight changes in road surface to approaching drivers/riders?				
5	Have signs and markers/studs been used to highlight locations where different layers of asphalt end, causing sudden changes in the road surface for approaching drivers/riders?				
F	Safety barriers				
1	Are safety barriers necessary and are they installed correctly?				
2	Have safe terminals (<i>that are suitable for the operating speeds through the work site</i>) been placed on the end of each section of barrier?				
G	Work site access				
1	Are work site access points located with adequate sight distances for entering and exiting work vehicles?				
2	Are all site access points closed off to unauthorised traffic?				
3	Are appropriate traffic controls in place where works traffic and public traffic interact?				
H	Road Worker Safety				
1	Do workers and supervisors at the works site wear the correct PPE as per HSE norms and as per the requirement of the activity being performed?				
2	Is safe ladder access provided for workers at site to reach the working platform/area?				
3	Is the working platform for safe working provided with toe guards and railings to prevent falls?				
4	Is there a suitable safety zone in advance of and beside the work site?				

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Appendix A – Road Safety Audit Checklists

Checklist for Stage 3: Construction Stage

Sl. No.	Issues	Yes	No	NA	Comments
I	Vulnerable Road Users (pedestrians, cyclists, motor cyclists and NMT) through the Work Site				
1	Are pedestrians, cyclists, motor cyclists, and NMT able to move safely through the construction zones?				
2	Are VRU able to cross the road safely?				
3	In particular, is safe access across the road provided for the young, the elderly and the disabled?				
4	Have all-weather footways been provided to assist public transport users to reach their stops without walking through the roadwork site?				
5	Do the construction layouts and practices allow adequate visibility for an approaching driver to see a pedestrian waiting to cross the road?				
6	Do the construction layouts and practices provide an adequate visibility funnel for an approaching driver to see a small vehicle waiting to cross the carriageway from a side road or a storage space of the median?				
J	Night Safety				
1	Are the road works conspicuous to all road users after dark?				
2	Are travel paths obvious to all road users after dark?				
3	Do carriageway closures have suitable advance warning zones and transition zones that are conspicuous and clearly delineated for night time?				

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Appendix A – Road Safety Audit Checklists

Table A.4 Checklist for Stage 4: Pre-Opening

Checklist for Stage 4: Pre-Opening					
Sl. No.	Issues	Yes	No	NA	Comments
A	General				
1	Has an RSA been carried out earlier?				
2	Have all earlier RSA recommendations been followed?				
3	Have all safety implications due to construction activity been addressed?				
4	Has the design been correctly translated into physical form?				
5	Have roadside hazards been cleared?				
6	Can the roadway be used safely by all users?				
B	New/Existing Road Interface				
1	Is the interface between the new road and the old road as safe as practical?				
2	Where drivers/riders depart the new road and return to the existing network, are they provided with sufficient delineation and pavement markings to compensate for possible increased speeds?				
C	Alignment				
1	Is the road safe to travel in both directions?				
2	Is the road safe to travel at night-time?				
3	Is there adequate visibility/stopping sight distance?				
4	Is the road free of any visual deceit?				
5	Is the transition from the old, unimproved road to the new road safe for users?				
D	Intersections				
1	Are sight lines free of obstructions at all intersections?				
2	Are the intersection layouts clear and visible from all approaches?				
E	Traffic signals				
1	Are traffic signals conspicuous, and functioning correctly and safely?				
2	Is the alignment of the signal head, correct?				
3	Are the signal lamps bright enough to be seen in day light?				
4	Are the signal timings reasonable and likely to maximise driver/rider compliance?				
5	Is the signal phasing reasonable and safe, with no conflicting movements?				
6	Are pedestrian signals installed at each signalised intersection and are they easy to see, with adequate WALK time and clearance time?				

Checklist for Stage 4: Pre-Opening

Sl. No.	Issues	Yes	No	NA	Comments
F	Roundabouts				
1	Are roundabouts visible and recognisable from all approaches?				
2	Are correct advanced warning and suitable direction signs installed on each approach to each roundabout?				
3	At other intersections are Stop or Give Way signs conspicuous and correctly installed to make priority clear?				
G	Interchanges				
1	Are sight lines open and free of obstruction at all merges and diverges?				
2	Are the distances between decision-making points sufficient for safe movement?				
3	Are the direction signs for each interchange clear and easily understood at the anticipated operating speeds?				
H	Signs, Pavement Markings and Delineation				
1	Do all signs satisfy the 6Cs of good signage practice? (<i>Conspicuous, Clear, Comprehensible, Credible, Consistent, and Correct</i>)				
2	Has the required number of signs been provided to warn, inform, guide, control or delineate?				
3	Have all unnecessary old signs and markings (<i>including those from the road works</i>) been removed?				
4	Are the speed zones provided with clear speed restriction signs of a suitable value?				
5	Are pavement markings continuous, correct, and conspicuous?				
6	Have the road signs and road markings been tested for retro-reflectivity?				
7	Have potentially confusing situations been removed or resolved safely?				
8	Are guideposts correctly installed in accordance with the layouts shown in the standard drawings?				
9	Are chevron alignment markers installed where necessary and in accordance with the layouts shown in the standard drawings?				
10	Is there consistency and uniformity of delineation and markings throughout the project?				
11	Have initiatives been taken to delineate the road and to maximise the chance that vehicles will remain on the road?				
I	Roadside Hazards				
1	Are all roadside hazards adequately treated? <i>Note: Hazards removed, relocated, or adequately marked?</i>				
2	Is a safety barrier installed where necessary?				
3	Is the barrier correctly and safely installed?				
4	Are impact attenuators provided in the gore area(s)?				

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Appendix A – Road Safety Audit Checklists

Checklist for Stage 4: Pre-Opening

Sl. No.	Issues	Yes	No	NA	Comments
5	Are all bridge abutments and culverts safely treated?				
6	Have large trees, boulders, etc. been sufficiently treated to improve roadside safety?				
7	Have the trees and other vegetation that obstruct driver and pedestrian sight lines been removed/relocated/treated?				
8	Is there adequate delineation/visibility of barriers and fences during night time?				
9	Has thick vegetation by the roadside been put far enough back from the edge of the pavement to enable a driver to take protective steps in time if any human or animal should run across the road from behind or within the vegetation?				
10	Have all sharp edged or pointed fixtures or tops of supporting verticals on the median crash barriers or on dividers (<i>which can hurt a motor cyclist in case of a collision or crash or loss of balance</i>) been removed?				
11	Have all village name boards or direction boards by the roadside with pointed ends (<i>that could hurt a passenger in a bus on window seat</i>) been removed?				
12	Bridges/Canal crossings: Are the open spaces by the side of ends of parapets covered by protective crash barriers or walls to prevent vehicles going into the river or canal?				
13	Has vegetation been removed, and paving provided for a length of 120 m at the median openings for U-turns or for pedestrians to cross and to provide clear visibility?				
14	Is the height of vegetation in the median less than 600 mm on curves?				
15	Is the median clear of any trees with trunk diameter greater than 100 mm? If not, are such locations enveloped by protective crash barriers?				
16	Are fixing details of pipe railings such that the entire length of pipe is smooth and continuous without any projection on the roadway side?				
17	Are entrances to abandoned roads properly fenced off?				
J	Vulnerable Road Users (pedestrians, cyclists, motor cyclists, NMT)				
1	Do all VRU have connectivity along the road, with suitable lateral clearance to motor traffic?				
2	Will pedestrians (<i>particularly the young, old and disabled</i>) be able to safely walk along the road?				
3	Are pedestrian crossing places marked, and are the pedestrians channelled to these crossings?				
4	Is the new road free of 'squeeze' points where VRU may be exposed to traffic?				

Checklist for Stage 4: Pre-Opening

Sl. No.	Issues	Yes	No	NA	Comments
5	Will pedestrians (<i>particularly the young, old and disabled</i>) be able to safely cross the road?				
6	Are dropped kerbs provided at all intersections and mid-block locations where pedestrians are to cross?				
7	Is the number and placement of the pedestrian facilities adequate and safe for the situation and the pedestrian numbers?				
8	Is each pedestrian facility clearly marked and conspicuous on each approach?				
9	Are the correct signs and pavement markings installed at each pedestrian facility?				
10	Is each crossing facility well illuminated at night so that pedestrians can be seen by drivers/riders?				
11	Do all mid-block traffic signals have pedestrian push buttons to allow pedestrians to register their demand to cross the road?				
12	Are medians in urban areas free of barriers, suitably wide and with a suitable surface to act as an effective pedestrian refuge?				
13	Are pedestrian paths provided through medians to permit pedestrians to cross 'at road level' and to assist disabled pedestrians?				
14	Is adequate safe parking and stopping provided for NMT?				
15	Are bus stops located where passengers will use them?				
16	Are bus stops well delineated and lit?				
17	Are bus stops sealed for all-weather use?				
18	Does the newly constructed road allow adequate visibility for an approaching driver to see a pedestrian waiting to cross the road?				
19	Does the newly constructed road provide an adequate visibility funnel for an approaching driver to see a vehicle waiting to cross the carriageway from a side road or a storage space of the median?				
20	Is there sufficient pedestrian guard railing?				
21	Has it been installed correctly?				
22	Is the cycle lane/way safe from motor vehicles?				
K	Drainage				
1	Does the new road appear to have sufficient cross fall or camber and adequate drainage to prevent ponding of rainwater from occurring?				
2	Are all drains safe (<i>covered, underground or outside the clear zone</i>)?				
3	Is the discharge from the drains at proper locations to prevent wash away of the road/rail line?				

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Appendix A – Road Safety Audit Checklists

Checklist for Stage 4: Pre-Opening

Sl. No.	Issues	Yes	No	NA	Comments
L	Safety barrier				
1	Are safety barriers provided everywhere they are needed?				
2	Are they long enough to prevent an out-of-control vehicle from reaching the hazard?				
3	Are the terminal arrangements (<i>upstream and downstream ends</i>) safe?				
4	Are the safety barriers placed in a way that they do not restrict visibility?				
5	Are the safety barriers placed in a way that they do not block pedestrian desire lines?				
6	Has a steel beam guardrail been designed and installed correctly? <i>Note: check beam height, post spacing, lateral clearance, spacer blocks, nuts and bolts, reflectors, terminal pieces, and whether the beams have been overlapped correctly.</i>				
7	Are the transitions between barrier types (e.g. <i>steel beam guardrail to concrete bridge parapet</i>) safe?				
8	Have all features that may create a safety problem been addressed?				
M	Bridges				
1	Is the cross-section of the approach road maintained across the bridge?				
2	Can pedestrians walk along the bridge safely?				
3	Is the bridge parapet safe (<i>height, design, level of containment, limit on size of openings, etc.</i>)?				
4	Are the ends of the bridge parapet adequately protected?				
N	Landscaping				
1	Is the landscaping 'safe' in terms of roadside hazards? <i>Note: Any tree with a trunk diameter of more than 100 mm is considered a roadside hazard, if planted within the clear zone.</i>				
2	Does the landscaping permit clear sight lines at all intersections, pedestrian facilities, and median openings?				
O	Lighting and night time issues				
1	Are all signs easy to see and read at night?				
2	Are the critical locations (<i>intersections, pedestrian facilities, bus bays, bus stops, truck lay-by, toll plaza, etc.</i>) conspicuous at night?				
3	Is lighting provided on road sections passing through built-up areas, service roads, above and below the grade separator, underpass, etc.				
4	Are the illumination levels of an appropriate standard, consistent with the needs of the location, pedestrians, and other factors?				
5	Is the 'through route' well signed, line marked and obvious to road users at night?				

Checklist for Stage 4: Pre-Opening

Sl. No.	Issues	Yes	No	NA	Comments
6	Is the new road free of visual deceit for road users at night?				
7	Are the lighting columns frangible? If not, are they outside the clear zone or protected by a safety barrier?				
8	Are all lighting poles in the median (<i>less than 2 m wide</i>) protected by safety barriers?				
P	Access				
1	Are all accesses to/from adjoining properties conspicuous and as 'safe' as practical?				
2	Is adequate fencing provided to keep animals off the road, especially in rural areas?				
Q	Services				
1	Are access chambers, lines, boxes, lighting columns etc. located in a safe place? (<i>i.e. clear of traffic lanes and behind a safety barrier</i>).				
2	Is there a safe place for maintenance vehicles to stop?				
R	Surface treatment				
1	Does the surface appear to be skid resistant?				
2	Is the surface clear of excessive bleeding of bitumen?				
S	Environmental				
1	Is planting located to avoid obstruction to visibility and sight lines?				
2	Has planting been done in such a way that it does not cause problems when mature (<i>i.e. size of trunk or canopy spread</i>)?				
3	Has planting been done in such a way that it will not obscure pedestrian movements near the edge of the road?				
4	Have all natural features that create a danger by their presence or effect on visibility, been removed or relocated?				
T	General Road Safety Considerations				
1	Is the new road as safe as practical given the local weather (<i>sunrise, sunset, snow, fog, dust storms, or rainstorms</i>)?				
2	Is the project free of all 'visual deceit'?				
3	Is the road surface free of gravel and sand, and with good skid resistance?				
4	Have the batter slopes of cuttings been treated to minimise the risk of rocks falling onto the new road?				
5	Have all the temporary arrangements, signs, etc. been removed and replaced by permanent arrangements?				
6	Has the proposal taken safe account of animals on the road, including livestock and wild animals crossing from adjacent forests or fields?				

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Appendix A – Road Safety Audit Checklists

Table A.5 Checklist for Stage 4: Pre-Opening

Checklist for Stage 5: Completion and Existing Roads					
Sl. No.	Issues	Yes	No	NA	Comments
A	General				
1	Has an RSA been carried out earlier?				
2	Have all earlier RSA recommendations been followed?				
3	Is the road section devoid of any recurrent safety concern?				
4	Is the road section devoid of any Police concern over frequent crashes?				
5	Is the road section devoid of any confusion between the road and the adjacent network?				
6	If a service road is present does the service road operate safely?				
7	Is there any problem with headlight glare?				
8	Are the road users safe from any change in traffic pattern due to development near the road?				
9	Is the surface of the road free from defects that may result in safety problems (<i>i.e. loss of control or skidding</i>)?				
B	Sight Distances				
1	Are all sight distances adequate for the speed of traffic using this road?				
2	Are safe overtaking opportunities provided?				
3	Are U-turn provisions conspicuous and 'safe'?				
C	Alignment				
1	Is the horizontal and vertical alignment suitable for the 85 th percentile speed of traffic?				
2	Are there sufficient clear overtaking sections?				
3	Have the sections with marginal visibility for overtaking (<i>dilemma zones</i>) been handled by signing and marking?				
4	Have the sections with sharp curves, after long straight sections, sharp curves after crests, or long downgrades been handled with proper signing and marking?				
5	Are super-elevation, transitions and carriageway widths sufficient for the ambient speed?				
6	Have long or steep hills been provided with climbing lanes to prevent unsafe overtaking?				
D	Intersections				
1	Are junctions located in safe places? (<i>Check in relation to horizontal and vertical alignments.</i>)				
2	Is the layout of junctions obvious on each approach?				
3	Does the layout accommodate all types of vehicles?				
4	Is the visibility from the side road adequate?				
5	Is the method of control appropriate? (<i>Priority/ signalled</i>)				

Checklist for Stage 5: Completion and Existing Roads

Sl. No.	Issues	Yes	No	NA	Comments
6	Are dedicated turning lanes adequate (<i>i.e. lengths and widths</i>) to accommodate the volume and mix of traffic?				
7	Is the signing on the approach to a junction adequate?				
8	Is there adequate provision for pedestrians and cyclists?				
E	Roundabout				
1	Is it designed to control speeds to 50 km/h?				
F	Traffic signals				
1	Do they operate correctly?				
2	Are they clearly visible (<i>in all conditions</i>)?				
3	Can signals be seen only by those who should see them?				
4	Are control boxes located in a safe place?				
G	Interchanges				
1	Are sight lines open and free of obstruction at all merges and diverges?				
2	Are the distances between decision-making points sufficient for safety at the operating speed?				
3	Is the direction sign for each interchange clear and easily understood at the operating speed?				
H	Cross Sections				
1	Are lane widths, shoulder widths and bridge widths, 'safe' for the traffic volume and mix?				
2	Are medians and islands of adequate width for the safety of likely users?				
3	Are the shoulders suitable for use by all vehicles and road users, including pedestrians, cyclists and NMT?				
4	Is appropriate super-elevation and extra width of carriageway provided on curves?				
5	Has pavement edge-drop been handled safely? <i>Note: Edge drop is the vertical level difference between the pavement surface and the adjacent shoulder surface.</i>				
6	Does the cross-section change with different speed limits?				
I	Roadside Hazards				
1	Are all large (<i>more than 100 mm diameter</i>) sign supports located outside the clear zone if they are not frangible?				
2	Are the shoulders of an appropriate design (<i>width, profile, surfacing, etc.</i>)?				
3	Is the clear zone free of obstructions or other hazards?				
4	If hazards cannot be removed, are they protected by safety barriers?				
5	Are all crash barriers correctly and safely installed?				

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Appendix A – Road Safety Audit Checklists

Checklist for Stage 5: Completion and Existing Roads

Sl. No.	Issues	Yes	No	NA	Comments
J	Safety Barrier				
1	Are safety barriers provided where necessary?				
2	Are they long enough to prevent an out-of-control vehicle from reaching the hazard?				
3	Are the terminal arrangements (<i>upstream and downstream ends</i>) safe?				
4	Are the safety barriers placed in a way that they do not restrict visibility?				
5	Are the safety barriers placed in a way that they do not block pedestrian desire lines?				
6	Has the steel beam guardrail been designed and installed correctly? <i>Note: check beam height, post spacing, lateral clearance, spacer blocks, nuts and bolts, reflectors, terminal pieces, anchorage and whether the beams have been overlapped correctly.</i>				
7	Are the transitions between barrier types (e.g. <i>steel beam guardrail to concrete bridge parapet</i>) safe?				
8	Have all features that may create a safety problem been addressed?				
K	Drainage				
1	Are adequate drainage facilities provided?				
2	Are all drains, culverts outside the clear zone covered, or behind a suitable barrier?				
3	Have the side drains been constructed to a safe standard for vehicles and pedestrians?				
L	Signs, Pavement Markings and Delineation				
1	Do all signs and pavement markings satisfy the 6Cs of good signage and pavement marking practice? (<i>Conspicuous, Clear, Comprehensible, Credible, Consistent, and Correct</i>)				
2	Is the speed zone 'safe', and clearly signed?				
3	Are pavement markings conspicuous and continuous?				
3	Are road signs and road markings tested for retro-reflectivity?				
4	Is the road well delineated, with warning signs, plastic guideposts, and chevron alignment markers installed as necessary and spaced properly?				
5	Is there a sufficient number of signs to warn, inform, guide, control or delineate?				
6	Are the signs legible?				
7	Are they located in a safe place?				
8	Do they give the correct message?				
9	Can the messages be clearly understood?				
10	Are all posts protected?				
11	Is edge delineation adequate?				
12	Are road markings correct and in good condition?				

Checklist for Stage 5: Completion and Existing Roads

Sl. No.	Issues	Yes	No	NA	Comments
M	Vulnerable Road Users (pedestrians, cyclists, motor cyclists, NMT)				
1	Do all VRU have connectivity along the road, with suitable lateral clearance to motor traffic?				
2	Will pedestrians (<i>particularly the young, old and disabled</i>) be able to safely walk along the road?				
3	Are pedestrian crossing places marked, and are the pedestrians channelled to these crossings?				
4	Is the new road free of 'squeeze' points where vulnerable road users may be exposed to traffic?				
5	Will pedestrians (<i>particularly the young, old and disabled</i>) be able to safely cross the road?				
6	Are dropped kerbs provided at all intersections and mid-block locations where pedestrians are to cross?				
7	Is the number and placement of the pedestrian facilities adequate and safe for the situation and the pedestrian numbers?				
8	Is each pedestrian facility clearly marked and conspicuous on each approach?				
9	Are the correct signs and pavement markings installed at each pedestrian facility?				
10	Is each crossing facility well illuminated at night so that pedestrians can be seen by drivers/riders?				
11	Do all mid-block traffic signals have pedestrian push buttons to allow pedestrians to register their demand to cross the road?				
12	Are medians in urban areas free of barriers, suitably wide and with a suitable surface to act as an effective pedestrian refuge?				
13	Are pedestrian paths provided through medians to permit pedestrians to cross 'at road level' and to assist disabled pedestrians?				
14	Is adequate safe parking and stopping provided for NMT?				
15	Are bus stops located where passengers will use them?				
16	Are bus stops well delineated and lit?				
17	Are bus stops sealed for all-weather use?				
18	Does the newly constructed road allow adequate visibility for an approaching driver to see a pedestrian waiting to cross the road?				
19	Does the newly constructed road provide an adequate visibility funnel for an approaching driver to see a vehicle waiting to cross the carriageway from a side road or a storage space of the median?				
20	Is there sufficient pedestrian guard railing?				
21	Has it been installed correctly?				
22	Is the cycle lane/way safe from motor vehicles?				

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Appendix A – Road Safety Audit Checklists

Checklist for Stage 5: Completion and Existing Roads

Sl. No.	Issues	Yes	No	NA	Comments
23	Are there adequate, safe facilities for pedestrian movement?				
24	Are pedestrian facilities used as intended?				
25	Have all locations of risk for pedestrians and NMT been eliminated?				
N	Speed management				
1	Does the geometric design (<i>cross-section, alignment, etc.</i>) reinforce the speed limit?				
2	Is the ambient traffic speed within speed limit?				
3	Is there a need for speed management measures?				
4	Have speed humps and other speed control devices of a safe design been installed?				
5	Are the devices adequately signed and marked?				
O	Access				
1	Are all accesses to/from adjoining properties conspicuous and as 'safe' as practical?				
2	Is adequate fencing provided to keep animals off the road, especially in rural areas?				
P	Bridges				
1	Is the cross-section of the approach road maintained across the bridge?				
2	Can pedestrians walk along the bridge safely?				
3	Is the bridge parapet safe (<i>height, design, level of containment, limit on size of openings, etc.</i>)?				
4	Are the ends of the bridge parapet adequately protected?				
Q	Lighting and night time issues				
1	Are all signs easy to see and read at night?				
2	Are the critical locations (<i>intersections, pedestrian facilities, bus bays, bus stops, truck lay-bys, toll plazas etc.</i>) conspicuous at night?				
3	Is lighting provided on road sections passing through built up areas, service roads, above and below the grade separator, underpass, etc.				
4	Are the illumination levels of an appropriate standard, consistent with the needs of the location, pedestrian and other factors?				
5	Is the 'through route' well signed, line marked and obvious to road users at night?				
6	Is the new road free of visual deceit for road users at night?				
7	Are the lighting columns frangible? If not, are they outside the clear zone or protected by a safety barrier?				
8	Are all lighting poles in the median (<i>less than 2 m wide</i>) protected by safety barriers?				

Checklist for Stage 5: Completion and Existing Roads

Sl. No.	Issues	Yes	No	NA	Comments
R	Environmental				
1	Is planting located to avoid obstruction to visibility and sight lines?				
2	Has planting been done in such a way that it does not cause problems when mature (<i>i.e. size of trunk or canopy spread</i>)?				
3	Has planting been done in such a way that it does not obscure pedestrian movements near the edge of the road?				
4	Have the natural features that create a danger by their presence or effect on visibility been removed or relocated?				
S	General Road Safety Considerations				
1	Is the new road as safe as practical given the local weather (<i>sunrise, sunset, snow, fog, dust storms, or rainstorms</i>)?				
2	Is the project free of all 'visual deceit'?				
3	Is the road surface free of gravel and sand, and with good skid resistance?				
4	Have the batter slopes of cuttings been treated to minimise the risk of rocks falling onto the new road?				
5	Have all the temporary arrangements, signs, etc. been removed and replaced by permanent arrangements?				
6	Has the proposal taken safe account of animals on the road, including livestock and wild animals crossing from adjacent forests or fields?				

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Appendix A – Road Safety Audit Checklists

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Appendix A – Road Safety Audit Checklists

Appendix B – Safety Audit Case Studies and Reporting

B.1 Case Study 1: Feasibility / Preliminary Design Stage Audit

The observations and recommendations are based on the Preliminary Design Stage Audit of highway projects conducted by the National Transport and Safety Authority. One issue and recommendation are presented as an example.

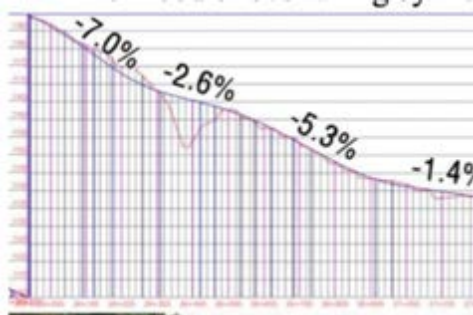
Table B.1 Safety Concerns and Recommendations at Preliminary Engineering Stage

No.	Safety Concerns & Audit Findings		Recommendations		Client Response
	Description	Risk	Description	Priority	
1	Road alignment and cross section				
	 At three places the road alignment crosses railway line. At two locations, the road passes through a underpass structure. At one location the road passes on a bridge. No drawings provided for crossing. There is a village near the bridge. Pedestrians may use the crossing. Pedestrians – vehicle collision may happen during operation. Construction time accident may happen.	Medium	• Provide a service road for the section. • Provide safety bar at rail crossing. • Provide guard rail for pedestrians. • Provide traffic calming measures to reduce speed at the rail junctions.	Desirable	

B.2 Case Study 2: Detailed Design Stage Audit

The observations and recommendations are based on the Detailed Design Stage Audit conducted for Kenya National Highways Authority. One issue and recommendation are presented as an example.

Table B.2 Detailed Design Stage Audit

No.	Safety Concerns & Audit Findings		Recommendations		Client Response
	Description	Risk	Description	Priority	
1	Consecutive downgrade profile				
	 There are several sections with consecutive downgrade profile. The driving speed increases on downhill profile. This can result in a crash. On the reverse side, continuous uphill gradient may cause engine failure. Slow upward movement by large vehicles may induce smaller vehicles to overtake and cause a risky situation.	Medium	Provide climbing lane. Provide parking area for breakdown vehicles. Provide transverse marking and rumble strips along with signs to advise speed reduction.	Desirable	

B.3 Case Study 3: Construction Stage Audit

The observations and recommendations are based on the Construction Stage Audit for widening of National Highways for the National Highways Authority of India. One issue and recommendation are presented as an example.

Table B.3 Safety Concerns and Recommendations at Construction Stage

No.	Safety Concerns & Audit Findings		Recommendations		Client Response
	Description	Risk	Description	Priority	
1	Temporary traffic diversion plan				
	Temporary traffic diversion plan has been submitted. The plan is too small to clearly understand. All features of the plan are not observed on the ground. Transition from two-way movement to one-way movement and vice versa are not clear. Management of pedestrians and other VRU not shown.	High	Need to present more legible drawings and get approval. Need to update traffic management plan as the work front shifts. Need to adhere to the plan with the use of proper safety material. Two – way traffic delineation is to be shown clearly. Movement from one-way to two-way and vice versa needs to be clearly shown. Area for pedestrian and other VRU movement should be clearly marked in work zones.	Essential	

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B.4 Case Study 4: Pre-opening Stage Audit

The observations and recommendations are based on the Pre-Opening Stage Audit of widening of a National Highway for the National Highways Authority of India. One issue and recommendation are presented as an example.

Table B.4 Safety Concerns and Recommendations at Pre-Opening Stage

No.	Safety Concerns & Audit Findings		Recommendations		Client Response
	Description	Risk	Description	Priority	
1	Vulnerable road users (pedestrians, cyclists, two-wheelers, animal drawn vehicles)				
	No signing, marking, or safe passage seen in the work zone for pedestrians / VRU. On the existing ROB, pedestrians walk at day and night time. They are not clearly visible. 	High	Provide facility for VRU. Provide pedestrian crossing signing and marking. Provide crossing at bus shelter locations with accessible median. Cover gaps in drainage or protect the area so that pedestrians do not fall down at night.	Essential	

B.5 Case Study 5: Post – Completion and Existing Roads Audit

The observations and recommendations are based on experience of highway construction in India and the existing stage road assessments carried out by the Kenya National Highway Authority. Two issues and recommendations are presented as an example.

Table B.5 Safety Concerns and Recommendations at Post-completion Stage

No.	Safety Concerns & Audit Findings		Recommendations		Client Response
	Description	Risk	Description	Priority	
1	End treatment of railings / guard rails				
	The ends of culvert parapets are not protected. Only a hazard marker has been placed to warn traffic. Vehicles may hit the end of railing, causing serious injury to driver and passengers. 	Very High	Place energy attenuators at the end of railings / guard rails to prevent high-speed crashes. Hazard markers can also be placed.	Essential	

Table B.6 Safety Concerns and Recommendations at Existing Stage

No.	Safety Concerns & Audit Findings		Recommendations		Client Response
	Description	Risk	Description	Priority	
1	Junction to Mathai Supermarket				
	No road marking / pedestrian crossing present. Risk of pedestrian – vehicle collisions due to wrong movement of vehicles. 	High	Provide road marking with thermoplastic. Place retro-reflective information, direction and regulatory signs. Provide properly marked at-grade pedestrian crossing.	Highly desirable	

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Appendix B – Safety Audit Case Studies and Reporting

Appendix C – Crash Data for Road Safety Audits

C.1 Crash data for Road Safety Audits

A Road Safety Audit is a proactive way to address road safety issues. It anticipates the crashes that can occur due to deficiencies in infrastructure and takes action before they occur.

Crash analysis is a reactive way of addressing road safety issues. Data on crashes occurring on a stretch of road are gathered, causes and types of crashes are analysed, and remedies are suggested.

It helps a road safety auditor to understand the safety aspects of the road if some data on crashes over the last 1 – 3 years are available. These data typically include the number of crashes per month by type, number of persons killed, injured and seriously injured. These data are obtained from the nearest police station on the road stretch. If the data is available for the location of the crashes, it is convenient for the audit team to look at the specific sections in more details.

Crash data are helpful for carrying out an audit at the feasibility and design stages to understand the deficiencies in the current design. For a construction stage audit, the crash data indicates the type of improvement required in traffic management during construction. For audits on existing roads, the auditor can get an indication of how high the risk of a crash is. If the number of crashes over time goes down after implementation of an audit recommendations, it gives an indication that the right steps were taken to address the safety issues.

The data that are useful for the auditor are:

1. Number of crashes per year.
2. Severity of crashes.
3. VRU crashes.
4. Type of crash.
5. Cause of crashes.

The auditor should not focus entirely on the crash data. This may distract the auditor from looking at the other high-risk situations where crashes have not happened yet but may happen. Pie charts and bar charts of the data are a good way of analysing and reporting.

C.2 Traffic Volume Data

In addition to the crash data, the traffic volume on the road also gives an indication of the risk of crashes. Any available classified traffic volume data, turning movement data, and pedestrian volume data helps the audit team to understand the nature of the traffic on the road. Typically, traffic volume studies are not conducted as a part of the road safety audit, unless required as a part of the Terms of Reference. Available secondary data are usually referred to.

The volume data of importance are: Average Daily Traffic, peak hour traffic, composition of traffic, breakup of motorised and NMT, daily and peak hour pedestrian traffic along and across the road.

Refer to RDM **Vol 1, Part 2: Traffic Surveys**, for details on traffic volume data.

C.3 Traffic Signal Data

Data regarding traffic signals at intersections are useful, if available. These include data on:

1. Whether the signal is pre-timed, semi-actuated or fully actuated.
2. Green, amber, and red time and pedestrian signal time.
3. Vehicular and pedestrian interval and clearance time.

The data can be analysed to assess whether sufficient time is provided for the vulnerable road users to cross the intersection in a safe manner.

C.4 Traffic Speed Data

Spot speed data at the mid-block sections and at the approach legs of an intersections can be useful. These give the auditor an idea about the prevalent speed on the road facility and helps them to assess whether additional safety features are needed.

Appendix D – Safe System Approach

D.1 Introduction

The Audit team should consider a Safe System approach while auditing road projects. This approach is based on the premise that death and injury are unacceptable and are avoidable. It seeks to ensure that no road user is subject to kinetic energy exchange in a crash which will result in death or serious long-term disabling injury. Safe System principles represent a fundamental shift from traditional road safety thinking, reframing the way in which traffic safety is viewed and managed.

D.2 Principles

Safe System principles are:

1. People make mistakes that can lead to road crashes.
2. The human body has a limited physical ability to tolerate crash forces before harm occurs.
3. It is a shared responsibility of the designers, builders, managers and users of road facilities to prevent crashes and to provide post-crash care.
4. All parts of safety system must be strengthened to multiply their effects; if one part fails, the road users are still protected.

D.3 Elements and Application

Safer Infrastructure

Safety treatments can reduce crash likelihood. Some examples of high-performing infrastructure treatments are:

1. Well-designed roundabouts can reduce fatal crashes.
2. Grade separated pedestrian crossings reduce crashes.
3. Wire rope barriers (i.e., cable barrier systems) at the centre and edges of roads reduce fatal crashes.

All road users need to be considered when designing or upgrading road infrastructure. This includes:

1. The design of road infrastructure and the broader street environment should start with the needs of the most vulnerable users and then progress to the safety needs of the least vulnerable.
2. A road design and corridor planning exercise that progresses through the needs of pedestrians, cyclists, animal drawn carts, motorcycles, cars, trucks and buses, ensure that appropriate function, speed, road space allocation, and design features are incorporated to deliver the best safety outcomes.
3. This is not only a critical step in planning new roads but also for safety review and retrofitting of existing network.
4. Traffic management measures play a significant role in influencing the road safety situation.
5. In situations where the traffic mix is highly diverse, user compliance levels are usually low. The rules and regulations are often set by a transport department that is separate from the road authority, resulting in coordination challenges.
6. Restricting vehicle operation on certain roads, or time can improve safety.

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Role of Speed Management

The principle of safe travel speed underpins a Safe System approach. Critical speed threshold levels in traffic crashes differ depending upon the type of crash. Table D.1 presents the crash severity risk associated with travel speeds. The crash types examined are:

- Vehicle with pedestrian or other vulnerable road user.
- Vehicle side impact with a pole or tree
- Side impact between vehicles at intersections.
- Head on crashes between vehicles, and
- Single vehicle run off road crashes.

Table D.1 Survival Impact Speeds

Impact speeds above which chances of survival or avoiding serious injury decrease rapidly		
Crash Type	Impact Speed	Example
Car/Pedestrian or Cyclist	30 km/h	Where there is a mix of vulnerable road users and motor vehicle traffic
Car/motorcyclist		
Car/Car (Side impact)	50 km/h	Where there is a likelihood of side impact crashes (e.g. intersections or access points)
Car/Car (Head-on)	70 km/h	Where there is no separation between opposing traffic streams

Source: PIARC Road Safety Manual, from Tingvall & Haworth, 1999.

Table D.1 illustrates that for the given situations, even for a vehicle with the best currently available safety features, if fatalities are to be avoided, the road should not support travel speed above 50 to 70 km/h. If roadside hazards are protected with barriers and intersections are treated to reduce speeds to 50 km/h, the travel speeds on the road can be 70 km/h. The addition of median barriers would enable higher operating speeds. Where motorcycles are a large proportion of the traffic, lower speed limits, around 40 km/h, are necessary.

Speed management is at the centre of developing a safe road system. Where infrastructure safety cannot be improved in the foreseeable future and the road has a high crash risk, reduction of speed limits with appropriate enforcement, are a critical.

Lowering speed limit from 100 to 90 km/h may reduce mean speed by 4 to 5 km/h if there is a reasonable level of enforcement. Research shows that this causes a reduction of up to 20% in the fatalities (e.g. Sliogeris, 1992).

Role of Vehicles

Promotion of the benefits of vehicle safety features needs to be carried out by road safety agencies. Road safety agencies can encourage:

1. Corporate fleet operators to procure safer vehicles.
2. Elimination of inappropriate vehicle advertising that show high speed vehicles.
3. Movement from motorcycles to public transport.

Compliance and Human Error

Maximising road user compliance with road rules is an important issue. This requires legislative arrangements; data systems for vehicles, driver licensing and offences; enforcement; justice system support; offence processing and follow-up capacities.

Human error is an important contributor to fatal and serious crashes. Clear consistent guidance along a route is necessary to reduce uncertainty and indecision among drivers.

D.4 Safe System Implementation

Road safety agencies can implement effective practice through demonstration programmes. The programmes should include area-based projects involving all relevant agencies and some national level policy reviews.

D.5 Auditing for Safer Roads

When conducting an audit, some points that should be checked in drawings are:

1. Is the proposed cross section safe for the type of road?
2. Any special features to be considered in the horizontal and vertical alignments?
3. Will the new road be easily understood by all the road users?
4. Do any parts of the design present direct risk to any group of road users?
5. Are there any obvious roadside hazards?
6. Has safety of vulnerable road users been considered for movement along and across the road?
7. Will weather conditions present safety issues?
8. Is the road safe at night?

D.6 Technical Tips for Road Safety Audit: Warn, Inform, Guide, Control and Forgive

During every audit, the auditors should check whether the road users are warned, informed, guided, controlled, and forgiven for their mistake.

- **Warn** – by providing signs, pavement markings or rumble strips. The message should be clear and simple. Additional and repeated warnings are needed.
- **Information** - to be provided in small amounts. Drivers should not be overloaded with too many destination names. Direction signs and lane direction arrows are to be provided.
- **Guide** - additional guidance is needed at some places, at road work diversions and at sharp curves on hill roads.
- **Control** - is necessary for a safe road network. A roundabout is safer than an uncontrolled intersection.
- **Forgive** – by providing a clear zone. Roadside hazards should be removed, relocated or marked. Barriers should be the last option. If used, they must be installed with proper end treatment.

D.7 Technical Tips: Safety Elements in Junction Layouts

Geometric elements are critical for safe manoeuvring of vehicles in priority roundabout and signal-controlled junctions.

At a T intersection, a long deceleration lane encourages overtaking by through traffic. Tapering for right turn lane can be used.

Side road alignment should be perpendicular to the main road. This will encourage turning movements at a controlled speed. At T intersections, median opening should be reduced in length. It will stop high-speed turning. Vegetation, boards should be removed from the central median at the junction.

When side road traffic joins the main road, it has to wait for a gap in traffic flow on the main road. If the vertical profile of the minor road is not flat at the junction, drivers wait on a slope, and cannot see the traffic on the main road properly. The minor road should be made flat for a minimum 5 m before meeting the main road. If the proportion of goods vehicle is high, the desirable flat distance is 10 m.

At roundabouts, the central island should be positioned such that straight through movement cannot take place even on the extreme outer lane.

D.8 Technical Tips: Roadside Hazards

Roadside hazards are a major road safety risk. A five-step strategy for managing roadside hazards can guide a road safety audit team.

Table D.2 Survival Impact Speeds

Steps	Strategies and tips
Step 1 - Keep vehicles on the road.	- A combination of good geometrical design, good information and good maintenance is needed. - An early-stage audit can focus on geometric design and run-off-the-road risk. - In later audits, other safety options like tactile edge lines, warning signs, guideposts and chevron markers should be looked for in the drawings.
Step 2 - Remove fixed objects from within the clear zone of the road.	- Check if an appropriate clear zone has been provided. - Check if fixed objects, with a diameter of 100 mm or more, are in the clear zone. They should be removed, if possible.
Step 3 - Relocate hazards to locations outside the clear zone.	- If it is not possible to remove a fixed object, it should be relocated further away. - For example, it may be possible to relocate a lighting column 8 m from the road, but it will be less effective in lighting the road from there. - The team may recommend a 4 m relocation and a slip-based lighting column as a safer option.
Step 4 - Alter the hazard to reduce the severity of a collision.	- Safer forms of road furniture can be used. - Large signs can be placed on signposts. - Frangible lighting columns can be advised for use. - Drivable end walls for culverts can be suggested
Step 5 - Install crash barrier to shield errant vehicles from the hazard.	- When the first four steps are not feasible, installation of crash barriers is an option. - The purpose of a crash barrier is to protect the passengers of an errant vehicle from striking a roadside hazard. - A crash barrier is a fixed object and can cause injury when struck. - If used, the audit team will need to check that the barriers are correctly designed and installed. - Safe end treatments are important. Traffic attenuators can be provided to shield a hazard at the gore area. - The type of crash barrier to be used – flexible, semi-rigid or rigid, depends on the type of hazard and its distance from the edge of carriageway. - For details of crash barrier, End treatments refer to RDM 1.3.

D.9 Technical Tips: Road Signs & Pavement Marking

Too many signs should not be used. It should be evaluated whether signs are the best solution. If signs are recommended, they should be properly installed. Signs must be conspicuous, clearly seen, comprehensible, credible, consistent, correct, properly designed and placed.

Properly laid markings give a clear message and can influence driving behaviour. Edge line marking improves road safety. If an edge line is placed very close to the kerb, it gets covered by soil, dirt, and stagnant water. A shyness distance (distance between the edge of the kerb and the edge line) of 0.5 m from the vertical face of the kerb/edge is advisable.

On a sharp curve with restricted forward visibility a psychological median made of ladder hatching can prevent head-on collision.

Crashes can occur at locations where the pavement width reduces. Object marking can be used to alert the drivers of narrow pavements.

At a junction, a physical island enveloped by a marked island is a safe practice. The physical island on the minor road should be set back in line with edge of the paved shoulder. This prevents island-hit crashes. The width of the slip lane can be marked such that two vehicles cannot move parallelly while turning.

Refer to RDM **Volume 6 Part 1: Road Marking Manual** and **RDM Volume 6 Part 2: Traffic Signs Manual** for use of proper road signs and marking.

D.10 Technical Tips: Pedestrians & Cyclists

A pedestrian and cycle crossing phase should be provided at signalised crossings at busy junctions. Medians or refuge islands should be used to give spatial separation from vehicles. Overpasses and subways are not popular and should only be used as a last resort.

An audit team needs to consider the safety needs of all pedestrians, cyclists and in particular the safety needs of high-risk groups such as school-age children and elderly pedestrians, the disabled and intoxicated pedestrians.

Auditors should check to ensure that foot and cycle paths are wide enough for the expected volumes and are continuous. Proper ramps should be provided to cater to the elderly and disabled people. The audit should check whether kerb crossings are provided, and the walkways are clear of obstacles.

In rural areas, wherever there is considerable pedestrian movement (near villages, schools etc.) paved shoulders will give pedestrians their own space to walk. They also help cyclists, motorcyclists, and animal-drawn carts. Auditors should check the cross-section drawings to see if paved shoulders are proposed for such stretches.

A highway passing through a settlement should have space for pedestrian movement along and across the carriageway. Pedestrians should be able to cross the carriageway via the shortest distance. Medians should be wide enough to serve as a refuge space. Considering the mix of traffic on urban roads, lane width can be reduced to give proper width for the pedestrian ways.

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Appendix D – Safe System Approach

Appendix E – Draft Terms of Reference and Audit Brief Contents

Draft Terms of Reference

The following provides draft Terms of Reference for use by the Client when commissioning an RSA. Details of the project and specific issues to do with the management of the audit are to be inserted where shown.

Terms of Reference for a [insert name of road agency]

Road Safety Audit of [insert name of the road project]

Background

The [insert name of road agency] has developed a project to [insert a brief description of the type and location of the project] in order to provide improved capacity, traffic performance and safety for all road users.

The Task

The task for this assignment is to carry out a [insert stage name] stage road safety audit of the project [insert name of project] so that potential road safety problems can be identified, discussed and minimised before the project is completed. The audit shall be undertaken in accordance with the process detailed in the RSA manual.

Scope of Services

The scope of services required of the audit team shall include, but not necessarily limited to the following:

1. The audit shall be undertaken by an audit team of [insert number] auditors.
2. The Team Leader should be a Senior Road Safety Auditor, empanelled by a competent authority.
3. The Audit Team should attend a commencement meeting with the Engineer and designers to obtain full information about the project and to understand the background of the project. A review of all documents provided by the Engineer will be undertaken prior to inspecting the site and prior to finalising the audit report.
4. Inspections of the entire site is to be undertaken in day time and at night to get a thorough understanding of the existing traffic situation.
5. The Auditors should consult the appropriate checklist in the RSA manual. They should look at the safety needs of all road users, especially the vulnerable road users.
6. The Auditors should prepare a concise RSA report in an acceptable format.
7. The Audit report shall include a clear description of all the safety concerns identified. It shall contain practical recommendations for each safety concern.
8. The Audit Team Leader is to sign and submit the audit report to the Engineer.
9. The Audit Team Leader is to attend a Completion Meeting with the Engineer to answer questions about the audit findings and to discuss the audit recommendations and possible design changes and/or remedial treatments.

The following information shall be made available by the road authority to the RSA Team Leader for the purpose of conducting the audit: [insert the list of reports, drawings, data, etc.]

Qualifications and Experience

The audit services are to be provided by a team comprising [insert number, typically 3 members] road safety engineering specialists and they shall be road safety auditors. Sound knowledge of road safety engineering and practical experience in highway design and traffic engineering is required by the audit team.

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Required Inputs

[Adjust these requirements to suit the scale and complexity of the project]

The assignment is expected to take up to [30] person days, as follows:

[9] person days for reviewing the reports/drawings and attending the commencement meeting

[9] person days for inspecting the site (day and night time inspections are required)

[6] person days for preparing the road safety audit report.

[6] person days to attend the completion meeting. (This will normally be held within one month of the audit report being submitted.)

Reporting

The Audit Team Leader shall submit the completed and signed RSA report to the Engineer in electronic format by [write submission date for the audit report]

Any questions about the proposal or the audit are to be directed by the Audit Team Leader to [insert name of the responsible engineer] via telephone [insert number] or email [insert email address].

Table E.1 Audit Brief Contents

SL No.	Contents
1	General information
2	Stage of the audit (<i>e.g. preliminary design, pre-opening</i>)
3	Project location and description
4	Contact details of the client and audit teams
5	Project background
a.	List of relevant documents (<i>e.g. plans, drawings and visualisation</i>).
b.	List of previous audits.
c.	Safe Systems Assessments (SSAs) and corrective action reports.
d.	Key road and traffic characteristics (<i>e.g. volumes, speed environment and crash data</i>).
6	Project requirements
a.	A clear requirement that the audit should be carried out with a focus on Safe System principles.
b.	An instruction to carry out the audit in accordance with a recognised guidance document and/or local policy.
c.	On-site inspections to cover relevant road conditions and/or specific road user groups (<i>e.g. thematic audits</i>).
d.	Timeframe and milestones (<i>including provision for commencement and completion meetings</i>).
7	Specific considerations
a.	Out-of-scope items (<i>e.g. issues related to interface with adjacent land use and rail corridor, structural integrity, personal security, and network operation considerations</i>).
b.	Audit team composition and particular expertise (<i>e.g. additional expertise required in human factors or a vulnerable road user group</i>).

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Appendix E – Terms of Reference and Audit Brief Contents

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